

Research paper

A novel computational paradigm for scheduling of hybrid energy networks considering renewable uncertainty limitations

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ABSTRACT

Numerous developing countries are currently grappling with energy crises due to the current surge in global fossil fuels prices. This situation has prompted policymakers and cities administrator to urgently seek alternative power sources. Scientists worldwide are actively working on innovative computational methods to address the coordination challenges faced by hybrid power systems. Their objective is to optimize the utilization of environmentally friendly hybrid power sources, such as clean energy sources (CESs) like wind and solar power, within the framework of traditional hydro-thermal coordination. This research aims to examine how integrating CESs into conventional energy delivering hydro-thermal coordination model can minimize the significant costs associated with energy production. The proposed approach involves constructing a probabilistic model to represent the hybrid energy coordination problem. To handle uncertainties related to CESs, a technique known as point estimation is employed. This technique employs Weibull and Beta distribution functions to handle uncertain input variables associated with wind and solar power. The present study employs an optimization paradigm that integrates the black widow optimization (BWO) algorithm with the radial basis function neural network (RBFNN). This paradigm not only considers the uncertainties in the system but also adapts its parameters to attain an optimum balance among the exploration as well as exploitation phase. Within the context of advancing energy transition the outcomes of our simulations demonstrate a significant achievement. The incorporation of CESs into the hydro-thermal coordination problem yields a substantial 10 percent reduction in operational costs and an impressive 64 percent decline in emissions. The convergence curves of the proposed scheme reveal that the presented hybrid computing paradigm quickly converges to the optimal solution, attaining the best global optimum solution. Three case studies have been conducted, in Case Study-I, the best attained optimal cost and emissions are \$3.6e4/day and 2.8e4 lb/day, respectively, with photovoltaic and wind energy costs at \$1148/day and \$3055/day, respectively. For Case Study-II, a complex function, emissions of 15887.42 lb/day and an energy production cost of \$26072.89/day are achieved. For Case Study-III, the optimal cost of \$26558/day and emissions of 12932 lb/day are acquired using the proposed approach. This valuable research contribution holds particular significance for the economic empowerment of developing nations. By diminishing their reliance on imported fossil fuels for energy generation, it paves the way for enhanced self-sufficiency. Moreover, the findings provide a well-defined pathway towards fostering sustainable and environmentally friendly energy production within urban settings. Additionally, the research highlights the transition from fossil fuel-dependent industries to the establishment of sustainable urban economies.

1. Introduction

1.1. Motivation

The investigation and advocacy of energy transition in urban areas is important due to cities central role in global energy consumption and

greenhouse gas emissions (GHGs) (Chien et al., 2023). To achieve net-zero, energy facilities must be in place to handle temporary fluctuations in solar and wind energy. It is recommended that clean energy be supplied and that, in cases where green sources are inadequate to fulfil the need, energy be imported (Ahmed et al., 2023a). Furthermore,

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Nomenclature	
GHGs	Greenhouse gas emissions
CESs	Clean energy sources
HTSP	Hydro-thermal scheduling problem
HTSWP	Hydro-Thermal-Solar-Wind Planning
SCOF	sine cosine optimization framework
PET	Point-estimation technique
NH	Number of hydro units
NW	Number of wind units
NS	Number of solar units
NT	Number of thermal units
T	Hours
h_i	Penalty factor
F	Singular composite objective function
F_{Cost}	Thermal function
$F_{emission}$	Emission function
W_j	Coefficient of correlation
P_{kNH}	Hydro output power
h_{kj}	Hydro net head
F_{wind}	Wind cost function
F_{solar}	Solar cost function
NASA	National aeronautics and space administration
RBFNN	Radial basis function neural network
MLFNN	Multilayer feedforward neural network
SDGs	United nation sustainable development goals

accelerated population increase, inadequate power infrastructure, economic difficulties, and geopolitical tensions in developing nations such as Latin America, African and Asian regions (Farghali et al., 2023). Cities are experiencing a greater demand for energy as a result of global growing cities, which has pulled existing infrastructures and escalated ecological problems to fulfil the United Nation Sustainable Development Goals (SDG) such as SDG 13. Urban centres must use clean energy sources (CESs) to reduce global warming, improve air quality, and meet ecological objectives (Singh et al., 2023). This study aims to explore challenges related to transitioning energy sources in urban environments. Coordination of hybrid power infrastructure is a common optimization task in this industry. Solar and wind generation are intermittent, thus utility administrators must carefully incorporate stable energy sources. This compensates for power shortages and maximizes power utilization during peak production. Simultaneously, power dispatch facilities have a responsibility to reduce the costs associated with transitioning among multiple sources of power throughout the execution of this operation (Ahmed et al., 2022d).

The primary objective of this study is to develop a sophisticated optimization paradigm based on soft computing strategies. This study presents an approach that employs the black widow algorithm for optimization purposes. This approach is employed to tackle an intricate planning challenge that involves the integration of energy facilities for hydroelectricity, thermo-electric units, photovoltaic facilities, and wind energy facilities. This paradigm aims to overcome the challenges hindering the adoption of electrical power in urban settings while also promoting cost-effectiveness and ecologically sustainable practices. The primary aim of this study is to mitigate the expenses associated with electric power production as well as the release of GHG pollutants from thermal power plants within a certain period. Utilizing readily available and affordable energy sources like hydroelectric, wind, and solar power will help achieve this. Furthermore, zero carbon plans in green

integration promote the widespread use and incorporation of clean energy sources in order to establish a carbon-neutral or emissions-free power industry and mitigate climate change. The issue at hand might be characterized as a multifaceted and constrained energy optimization challenge. The findings of this study will make a valuable contribution to the advancement of urban planning, power management, and policy development. Furthermore, this will enhance the resilience and livability of cities for present and prospective residents.

1.2. Literature review

Table 1 provides an comparison overview of the literature survey pertaining to soft computing algorithms employed for addressing the energy dispatch problem with present work. For better use of solar energy in power networks, Stephen et al. Afonaa-Mensah et al. (2023) explored strategy to reduce variability in solar resources. A full day dynamic energy dispatch scheme was investigated for solar-load linkages affected solar energy reception and, by extension, production expense. Nevertheless, the cost model is the only optimization model taken into account in this work, and convex constraints are the only ones examined. Duan et al. (2023) introduced a distributed scheme to address the dynamic dispatch model for hybrid energy networks. They employed a weight-sum method in conjunction with the Lyapunov function approach to achieve the optimal solution for hybrid networks. Nevertheless, it is important to note that this study overlooked two significant non-convex constraints, namely, restricted operating regions and transmission losses. In the investigation cited as Basak et al. (2023), the researchers employed hybrid heuristic algorithms to achieve the optimal solution for operational costs in energy networks. The study utilized both the crow search and JAYA algorithms to address various network load scenarios, considering a linear profile for wind energy. It is important to note, however, that this research overlooked system-associated constraints and demonstrated suboptimal performance in addressing the uncertainties associated with green energy sources. Several other studies, including those referenced as Sheng et al. (2023), Sundaram (2022), have explored single and dual optimization functions for dispatch models to achieve the optimal solution for the generation cost of hybrid energy networks. The comparative analysis, as illustrated in Table 1, indicates that the model presented in this work is both more intricate and practical. This is attributed to its consideration of practical constraints and the development of an efficient approach to handle the uncertainties linked with CESs.

The optimization of electric power efficiency is a fundamental problem in the context of contemporary urban building projects, exerting a significant influence on the long-term viability and robustness of cities. Some other applications of optimization can be observed in waste management (Alruqi and Sharma, 2023), the synthesis of biohydrogen (Sharma et al., 2023a), and theoretical frameworks for chemical splittings (Sharma et al., 2023b). In recent times, scholars have employed a wide range of techniques to confront the challenge of engineering optimization in the domain of energy delivery and effective utilization (Zhang et al., 2023; Alzahrani et al., 2023). Recently, traditional convex optimization techniques are implemented to address a range of planning challenges. These strategies encompass linear programming, which utilizes computational frameworks to optimize linear objectives while adhering to linear limitations (Lauinger et al., 2016). Conversely, Lagrange relaxation entails the relaxation of limitations to simplify complex optimization issues in electric cars (Adacher et al., 2023). Furthermore, mixed-integer programming integrates both continuous and discrete decision variables in optimization models, which improves its adaptability in CESs consumption for distributed energy networks (Huang et al., 2022). Optimizing scheduling solutions also uses gradient search procedures, which are a type of iterative improvement based on the gradient of an objective function (Vijayalakshmi et al., 2022). These conventional approaches serve as fundamental tools for addressing the intricate task of optimizing schedules across various

Table 1
Comparison of power dispatch research in recent years considering CESs.

References	Published year	Types	Objectives	Constraints	Approach
Afonaa-Mensah et al. (2023)	2023	Thermal Solar	Cost Model	Valve-point loading Ramp-rate Limits Power balance	Particle swarm algorithm
Duan et al. (2023)	2023	Thermal Solar Wind	Cost Model Emission Model	Capacity Constraints Ramp-rate Limits Power balance	Distributed algorithm
Basak et al. (2023)	2022	Thermal Wind	Cost Model	Capacity Constraints Power balance	Crow search and JAYA algorithm
Sheng et al. (2023)	2023	Thermal Solar Wind	Cost Model Emission Model	Capacity Constraints Power balance	Improved COOT algorithm
Sundaram (2022)	2022	Thermal	Cost Model Emission Model	Capacity Constraints Power balance	Multi-verse algorithm
This work	–	Thermal Solar Wind Hydro	Cost Model Emission Model	Capacity Constraints Ramp-rate Limits Power balance Prohibited regions Spinning reserves Hydro	IBWOA-RBFNN

contexts. However, due to the complicated nature of decision variables, which are defined by the fact that they are not convex and are not linear, and the fact that the system has strict rules, the results from these methods are often inaccurate and inefficient.

In the past decade, a multitude of algorithmic evolution and computational techniques based on swarm intelligence have been employed for addressing traditional scheduling dilemmas and nonetheless, each of these methods has its own limitations and benefits ([Mishra and Kane, 2023](#)). For example, evolutionary optimization framework garnered significant attention as well as applause owing to their inherent adaptability and reliability in effectively discerning the most suitable options ([Abdelkader et al., 2022](#)). The application of evolutionary algorithms to tackle optimal planning challenges in various domains encompasses: urban planning ([Jiang et al., 2023](#)), aerospace engineering ([Periaux and Tuovinen, 2023](#)), power system operations ([Ahmed et al., 2022b](#)), buildings structural design ([Dixit and Stefańska, 2023](#)), and transportation ([Ahmed et al., 2023c](#)). Currently, there is a notable emphasis among academics on the utilization of population based intelligence-driven algorithms and meta-heuristic soft computational optimization approaches. This is due to their robustness and increased effectiveness in tackling complex multi-dimensional restrictions and real-world engineering problems. The utilization of these algorithms to address power coordination issues manifests are: artificial bee colony ([Hakli, 2023](#)), bacterial foraging optimization ([Li et al., 2022](#)), krill herd algorithm ([Hu et al., 2022](#)), and whale optimization ([Ahmed et al., 2022a](#)).

Meta-heuristic techniques offer a greater potential for circumventing local optima and uncovering the optimal global solution, and numerous studies from researchers worldwide have been documented concerning the power scheduling problems for sustainable urban power consumption. [Hammid et al. \(2023\)](#) performed a comprehensive analysis and comparison of several approaches utilized in dealing with the hydro-thermal scheduling problem (HTSP). In the modern era, there has been an increasing inclination towards proposing hybridized fundamental algorithms as a means to address the challenges posed by HTSP. An example of a hybrid approach was introduced by [Balachander et al. \(2022\)](#), which combines a bacterial foraging algorithm with a particle swarm technique to tackle the problem of HTSP. This approach takes into account the complex constraints that arise from the chronological interconnections of plants and the consequences of valve-point loading effect limitations. The authors of [Ghosh et al. \(2019\)](#) utilized the ABC-BAT algorithm in tandem to tackle the problem of the HTSP. The research conducted by the authors encompassed two unique

approaches: one strategy aimed at minimizing costs, while the other strategy focused on improving the general efficiency of the system. While hybrid methods indeed offer improved convergence, they fall short in terms of computational resource efficiency. Additionally, these hybrid strategies present greater implementation challenges compared to alternative approaches.

The global popularity of wind and solar energy is increasing owing to its cost-effectiveness and ecologically sustainable characteristics. In comparison to the situation a decade ago, there has been a notable increase in the percentage of worldwide energy consumption derived from CES, rising from 8% to 18.3% in the year 2015 ([Heaton et al., 2023](#)). Within the context of present-day urban planning, there is a compelling need to adopt a more strategic approach towards the utilization of CESs ([Khavarian-Garmsir et al., 2023](#)). This need arises not only from the desire to achieve cost efficiency but also from the aspiration to attain the lofty objective of completely eliminating carbon emissions ([Kasaeian et al., 2023](#)). The prevailing trend in power infrastructure development highlights the imminent transition towards the prioritization of clean energy consumption. The integration of green power systems into conventional power scheduling paradigms has been driven by a common worldwide concern shared by energy managers and urban planners ([Meraj et al., 2023](#)). Nevertheless, the incorporation of this integration adds a new level of complexity to existing energy models. The complexity associated with sustainable energy sources provide intrinsic limitations that standard optimization methods are not well suited to handle. Consequently, there exists a significant deficiency in the capacity of conventional optimization techniques to provide all-encompassing techno-economic solutions that are imperative for effectively addressing the complexities of the contemporary energy environment ([Chu et al., 2023](#)).

Upon doing a full analysis of the literature described above, it becomes apparent that there exists a noticeable gap. This gap presents an opportunity for the development and implementation of a composite energy planning model. The objective of this model is to function as a flexible and comprehensive solution that can effectively tackle the complex issues associated with energy production in several domains.

1.3. Contributions

The primary objective of this research is to develop a robust framework for Hydro-Thermal-Solar-Wind Planning (HTSWP) by employing

the adaptive crow optimization algorithm. Integrating CESs into hydro-thermal synergy reduces costs and promotes sustainability. This research seeks the most economical power production mix by optimizing the distribution of resources among traditional and green systems. Advanced modelling and simulation analyse hydro, thermal, and CES dynamics to improve system understanding and prediction accuracy. This research also addresses grid stability, market dynamics, risk assessment, environmental effect, and technology innovation for building a robust, dependable, and sustainable electricity network. The study places particular emphasis on the utilization of hybrid power systems to generate clean energy. The creation of this model possesses a double relevance as it enables particular cities to achieve techno-economic solutions while also contributing to the global effort of achieving a low-carbon and resilient future. Furthermore, the present study assumes a crucial role in promoting the adoption of sustainable energy strategies in urban environments, hence generating significant consequences for a wide range of interested parties, such as politicians, urban planners, industry leaders, and local communities. The major contributions of this study are as follows:

1. This study distinguishes itself from previous scholarly publications (Das et al., 2018; Yaşar and Özyön, 2020) by specifically addressing a new hydro-thermal scheduling problem that incorporates an extra aim related to thermal plant emissions. In order to enhance the genuineness and applicability of the model, this work integrates two supplementary constraints: spinning reserves and restricted operation zones, hence distinguishing it from prior research (Kumar and Dhillon, 2022; Dasgupta et al., 2020a). Implementing the HTSWP coordination idea improves the energy generation of eco-friendly power sources and simultaneously decreases greenhouse gas emissions from thermal power plants. This paradigm enables urban planners and energy managers to effectively accomplish their long-term objectives related to the environment, society, and finances.
2. The hybrid black widow search method has been suggested as a possible strategy for efficiently addressing the multi-dimensional coordination problem in HTSWP. This strategy showcases proficiency in efficiently handling intricate multi-dimensional constrained optimization issues, hence establishing itself as a novel and successful optimization technique. Compared to previous attempts (Dasgupta et al., 2022, 2020b), the suggested optimization framework exhibits enhanced adaptability, leading to more precise optimal solutions with reduced computational requirements. Furthermore, the sine cosine optimization framework (SCOF) used in prior research lacks robust adaptive capabilities to dynamically adjust its parameters in real-time during the optimization process.
3. This study introduces new ideas and innovative approach that attempt to speed up the transition to CESs and enhance energy utilization. A novel framework is suggested to enhance the modelling of uncertain energy dynamics of CESs. This framework integrates the point-estimation technique (PET) with RETScreen meteorological data. This distinctive attribute has the capacity to propel progress in urban design, energy management, and policy development, leading to the construction of more resilient and habitable cities for current and future generations.

The structure of this paper is outlined in following manner. Section 2 discusses the mathematical modelling of HTSWP. Section 3 discusses the PET for CESs uncertainty modelling. The HTSWP computing paradigm is presented in Section 4, while Section 5 provides details on the RETScreen software. Section 6 covers the simulation results. Finally, Section 7 presents the conclusion and future research directions.

2. HTSWP mathematical modelling

The main goal of the HTSWP initiative is to reduce operating costs related to energy generation and mitigate pollutants arising from the combustion of fuel in energy hubs. Furthermore, this paradigm facilitates the utilization of environmentally friendly and economically feasible options for energy generation for urban infrastructure.

This part presents the mathematical representation of the research's layout, with the goal of achieving power output equilibrium among energy production hubs, while taking into consideration system limitations. Every individual energy source is quantitatively classified based on its unique constraints during a certain time period. Let us examine a hybrid power network comprising NT thermo-electric generators, NH hydro-electric units, NS photovoltaic power modules, and NW wind generators. The hybrid configuration presented in this study is specifically engineered to function during a time period denoted as T hours. During this length, the configuration aims to optimize the generation of power schedules by effectively using carbon-free sources. The main goals of this design are to utilize the sustainable energy option while reducing running costs of power network and also minimize the discharge of GHGs. Moreover, the reservoir capacity of each hydro-electric unit, photovoltaic system and wind units are strategically adjusted to its utmost potential, guaranteeing that the generated power schedule seamlessly coincides with the pre-established urban load-demand needs.

2.1. Mathematical formulation of thermo-electric units

One of the key components of hybrid power networks is comprised of thermal units that are responsible for the distribution of electricity. The primary emphasis pertains to the maximization of operational efficiency and reduction of emissions in relation to these thermal units. The prioritization of this purpose is accomplished by the efficient utilization of diverse alternative environmentally sustainable resources. The mathematical formulation of the thermal cost and emission model is expressed as follows (Ahmed et al., 2022e):

$$F_{cost} = \sum_{k=1}^T \left(\sum_{i=1}^{NT} t_k (a_{1i} + b_{1i} P_{ki} + c_{1i} P_{ki}^2 + |f_{1i} \sin(e_{1i}(P_i^m - P_{ki}))|) \right) \quad (1)$$

$$F_{emission} = \sum_{k=1}^T \left(\sum_{i=1}^{NT} t_k (x_{2i} + y_{2i} P_{ki} + z_{2i} P_{ki}^2 + d_{2i} \exp((e_{2i} P_{ki}))) \right) \quad (2)$$

The variables a_{1i} , b_{1i} , c_{1i} , e_{1i} , and f_{1i} in Eq. (1) represent the coefficients related to fuel cost. The expression $|f_{1i} \sin(e_{1i}(P_i^m - P_{ki}))|$ represents the effect of valve-point loading. In this context, P_i^m and P_{ki} denote the minimum power and output power of unit i correspondingly, whereas t_k denotes a certain time interval. In Eq. (2), the variables x_{2i} , y_{2i} , z_{2i} , e_{2i} , and d_{2i} are utilized to represent the coefficients related to emissions from the thermal plant.

Taking into account the influence of the price penalty factor h_i , the dual objective functions described by Eqs. (1) and (2) can be unified into a singular composite objective function denoted as F . This amalgamated objective function is formulated as follows (Ahmed et al., 2023b):

$$F_{composite} = \sum_{k=1}^T \left(\sum_{i=1}^{NT} t_k (F_{cost(i)} + h_i (F_{emission(i)})) \right) \quad (3)$$

where h_i is the penalty factor and computed as follow (Lauinger et al., 2016):

$$h_i = \frac{F_{cost(i)(max)}}{F_{emission(i)(max)}} \quad (4)$$

Penalty factors h is used to handle errors or deviations from limits.

2.2. Mathematical formulation of hydro-electric units

Fluctuations in water outflow, which are contingent on the output power P_{kNH} and the net head h_{kj} , serve as key influencers in hydroelectric power generation. In this context, the Glimm–Kirchmayer model is employed to depict the subsequent expression for the flow rates during the k th sub-period of the j th hydro unit (Secui et al., 2020).

$$Q_{kj} = \begin{cases} \mathcal{W}_j \eta(P_{kl}); \text{Fixed head with a large reservoir,} \\ \mathcal{W}_j \eta(P_{kl}) \mu(H_{kj}); \text{Varying head with a short reservoir,} \\ (l = j + NT; j \in [1, NH]; k \in [1, T]). \end{cases} \quad (5)$$

Within Eq. (5), $\mathcal{W}_j$ stands for the coefficient of correlation. The terms $\eta(P_{kl})$ and $\mu(H_{kj})$ represent the features of hydro-generation and reservoir water head, respectively. These characteristics are subsequently elaborated upon as follows.

$$\eta(P_{kl}) = \alpha_j + \beta_j(P_{kl}) + \gamma_j(P_{kl}^2) \quad (l = j + NT; j \in [1, NH]; k \in [1, T]), \quad (6)$$

$$\mu(H_{kj}) = X_j + Y_j(P_{kl}) + Z_j(P_{kl}^2) \quad (j \in [1, NH]; k \in [1, T]), \quad (7)$$

Here, α_j , β_j , and γ_j denote the fluid discharge parameters associated with the j th hydro unit. Meanwhile, X_j , Y_j , and Z_j signify the changing levels of hydraulic head pertaining to the same j th hydro unit.

2.3. Mathematical formulation of photovoltaic units

The following model is used to demonstrate the expenditures associated with the generation of solar energy (Ahmed et al., 2022c).

$$F_{solar} = \sum_{k=1}^{NP} P_{S,k} \times \nabla_i \Xi_k. \quad (8)$$

In Eq. (8), F_{solar} and $P_{S,k}$ represent the cost related to photovoltaic operations and the amount of photovoltaic modules used, respectively. The energy produced by the photovoltaic module depends on three main factors: the sunlight intensity, temperature conditions, and the inherent characteristics of the module. The beta probability density function is employed in this study to describe the solar irradiation. The following equation depicts the arrangement of this distribution (Ahmed et al., 2024):

$$F_{pdf}(\rho) = \begin{cases} \frac{\xi(\mathcal{V}+\mathcal{K})}{\xi(\mathcal{V})\xi(\mathcal{K})} \times \rho^{\mathcal{V}-1}(1-\rho)^{\mathcal{K}-1} \\ \text{for } 0 \leq \rho \leq 1, \quad \mathcal{V} \geq 0, \mathcal{K} \geq 0 \\ 0, \text{ otherwise.} \end{cases} \quad (9)$$

In Eq. (9), the symbols $\mathcal{V}$ and $\mathcal{K}$ stand for the parameters of the beta probability density function, while ξ refers to the gamma function. This function can be articulated by utilizing the mean value δ and the standard deviation Υ .

$$\mathcal{V} = \delta \left(\frac{(\delta(\delta+1))}{\Upsilon^2} - 1 \right), \quad (10)$$

$$\mathcal{K} = (1-\delta) \left(\frac{(\delta(\delta+1))}{\Upsilon^2} - 1 \right). \quad (11)$$

The quantity of electricity generated by photovoltaic modules is primarily determined by two critical factors: the level of solar radiation and the temperature of the cells. These factors can be calculated using the equation provided below (Ahmed et al., 2023e)

$$P_s(t) = C\mathcal{N}_{sr} \times \mathcal{N}_{pr} [P_{stc} \times \frac{P_{rad}}{P_{rad,stc}} \times [1 - \chi \times (Q_{cell} - Q_{cell,stc})]]. \quad (12)$$

Within expression (12), $P_s(t)$ signifies the incident solar radiation on the photovoltaic modules over the time interval t . The quantities $C\mathcal{N}_{sr}$ and the notation $C\mathcal{N}_{pr}$ denotes the total number of cells coupled in

both parallel as well as series connections, respectively. In addition, P_{stc} and $P_{rad,stc}$ represent the solar energy produced and exposure to radiation determined under standardized conditions for testing. Q_{cell} and $Q_{cell,stc}$ refer to the module's temperature during operation as well as temperature at which it operates during conventional testing situations, respectively.

The temperature of the photovoltaic cell module, denoted as Q_{cell} , is determined using Eq. (13), which is expressed in the following manner:

$$Q_{cell} = Q_{ambient} + \frac{P_{stc}}{P_{rad,stc}} \times (Q_{normal,temp} - 20). \quad (13)$$

Within (13), the variables $Q_{normal,temp}$ and $Q_{ambient}$ represents the typical temperature during operation as well as the external temperature at the time of the photovoltaic panels, accordingly.

2.4. Mathematical formulation of wind units

The present model is used to demonstrate the expenditures associated with the generation of wind energy (Ahmed et al., 2022c).

$$F_{wind} = \sum_{j=1}^{NW} \sigma_{p,j} \times \Theta_{aj}, \quad (14)$$

In the equation labelled as (14), the symbols F_{wind} , NW , and $\sigma_{p,j}$ correspond to the cost of producing wind energy, the count of wind mills units, and the produced energy measured in megawatts (MW), respectively.

Varying wind speeds have a significant impact on the amount of power that can be generated by wind turbines. Several techniques have proven that wind velocity behaviour is intrinsically unpredictable. In this particular research, the Weibull probability density function is applied to express the properties associated with wind current and capture the probabilistic aspect of wind velocity situations. What follows is a description of the analytical form that this density function takes (Guo et al., 2021):

$$W_{pdf}(\psi_{CWV}) = \left\{ \frac{\Lambda}{\tau} \left(\frac{\psi}{\tau} \right)^{\Lambda-1} \cdot \exp \left(- \left(\frac{\psi}{\tau} \right)^{\Lambda} \right) \quad \psi > 0. \right. \quad (15)$$

Within (15), the symbols Λ and τ embody the characteristics of shape and scale, respectively. ψ_{CWV} signifies the prevailing wind velocity, measured in meters per second. Wind turbines' generated electricity is determined by the velocity-energy bond, which is illustrated below:

$$\Pi_p = \begin{cases} 0, & (\psi < \psi_{in} \text{ OR } \psi \geq \psi_{out}), \\ \frac{\mathfrak{F}_r(\psi - \psi_{in})}{\psi_r - \psi_{in}}, & (\psi_{in} \leq \psi \leq \psi_r), \\ \mathfrak{F}_r, & (\psi_r \leq \psi \leq \psi_{out}). \end{cases} \quad (16)$$

In the equation designated as (16), the variables σ_p and $\mathfrak{F}_r$ stand for the generated energy and the rated energy capacity of the turbine, respectively. The parameters ψ_{in} , ψ_{out} , and ψ_r signify the wind velocities associated with input, output, and the rated conditions, respectively.

2.5. Combined HTSWP model of hybrid power networks

The expenses related to hydro-electric plants are typically disregarded in the coordination problem of HTSWP, as they tend to be significantly lower when contrasted with the operational expense of thermo-electric, wind generators, and solar power production facilities. In order to formulate the combined HTSWP, the subsequent expression is employed.

$$F_{(HTSWP)} = F_{composite} + F_{solar} + F_{wind} \quad (17)$$

Following that, this study give the formulation of the complex optimization function, which incorporates all the operational cost functions associated with the hybrid power sources. The objective purpose of this

study is to propose a cost-effective and ecologically sustainable energy solution to meet the energy demands of a metropolitan area. Furthermore, this versatile function considers the distinct constraints associated with each power source present in the system. Consequently, the objective function of HTSWP becomes notably intricate due to the interplay of conflicting objectives and interconnected system limitations inherent to it.

$$TFC = \min(HTSWP), \quad (18)$$

Subject to Eq. (19) to Eq. (23).

2.5.1. Hybrid power sources system constraints

The contributing power plants in the HTSWP method management are required to adhere to a variety of generating limits, including both linear and nonlinear constraints. This subsection outlines a number of system limitations that have been considered for the purpose of this study.

1. **Power Balance Limitation:** The combined power output from the participating machines must balance out to match the total power demand while accounting for power losses. This relationship can be mathematically represented as follows (Ahmed et al., 2023d):

$$\sum_{i=1}^{NH} \sum_{t=1}^{NT} \sum_{s=1}^{NS} \sum_{w=1}^{NW} (P_i) = P^{loss} + P^{load}. \quad (19)$$

2. **Generating Machines Limitations:** Usually provided by the supplier, these restrictions relate to the physical characteristics of the producing machine. Mathematically, these restrictions can be formulated as follows:

$$P_{ki} = \begin{cases} P_{ki}, & (P_i^m \leq P_{ki} \leq P_i^M), \\ P_i^m, & (P_{ki} < P_i^m), k \in (1, T), \\ P_i^M, & (P_{ki} > P_i^M), \\ i \in (1, NH + NT). \end{cases} \quad (20)$$

3. **Water Accessibility Limitations:** Throughout the designated timeframe, there exists a confinement on the cumulative outflow (Q_{kj}) from each hydro unit, ensuring it does not exceed a predefined reservoir water volume (V_j). This can be articulated mathematically as follows (Ahmed et al., 2023d):

$$\sum_{k=1}^T t_k Q_{kj}(P_{kl}) - V_j = 0. \quad (21)$$

4. **Ramp-rate Limitations:** Any units that modify their operational output state in terms of megawatts per hour (MW/h) must adhere to a specific ramp rate. This condition can be mathematically represented as follows (El-Hawary and Ravindranath, 1988):

$$\max \{P_{ki}^m, P_{ki} - DR_i\} \leq P_{ki} \leq \min \{P_{ki}^M, P_{ki} + UR_i\}. \quad (22)$$

5. **Spinning Reserve:** The spinning reserves (SRS) allocated to each energy source are symbolized as S , while the maximum aggregate spinning reserves contributed by all power-generating units are represented by S_R . Mathematically, this relationship can be stated as follows:

$$\sum_{i=1}^{NH} \sum_{t=1}^{NT} \sum_{s=1}^{NS} \sum_{w=1}^{NW} (S_i) = S_R, Full. \quad (23)$$

6. **Equilibrium in the Water Dynamics of Hydro-units:** This term pertains to the state of stability and uniformity in the flow and allocation of water within the hydro storage. This entails managing the input and outlet of water, together with other factors that

affect water levels, in order to preserve the total equilibrium and operation of the hydro storage component (Secui et al., 2020).

$$V_{hjk}^T = V_{hjk}^{T-1} + I_{hjk}^T - Q_{hjk}^T - S_{hjk}^T + \sum_{n=1}^{NH} (Q_{hn}^{T-TD} + S_{hn}^{TL-TD}) \quad (24)$$

7. **Water Storage and Discharge Limitation:** These limitations are in place to guarantee the optimal operation of the basin and its corresponding power generation infrastructure (Dasgupta et al., 2022).

$$\begin{aligned} V_{hjk}^m &\leq V_{hjk} \leq V_{hjk}^M \\ Q_{hjk}^m &\leq Q_{hjk} \leq Q_{hjk}^M \end{aligned} \quad (25)$$

This model is notably intricate, offering a heightened level of practicality and multidimensional applicability for energy operators. It effectively incorporates the realistic constraints inherent in the operation of energy sources within modern urban energy distribution facilities, rendering it a more feasible solution.

3. PET for CESs uncertainty modelling

The notion of PET was first proposed by Emilio Rosenbluth, a Mexican engineer, in the year 1975. The objective of this study was to evaluate the probability of system breakdowns. The efficacy of PET as a probabilistic approach for addressing engineering challenges has been well-established. Following this, Harr put up a revised iteration of PET with the aim of streamlining the computational complexity associated with simulations. Despite its computing efficiency, Harr's technique, referred to as PET, had difficulties when handling symmetric variables. In order to overcome the constraints associated with PET using symmetric variables, Hong proposed a unique variation of PET that maintains computing efficiency while accepting parameters that are both asymmetric and symmetric. The PET method functions as an approximation approach by employing a collection of variables to provide a singular solution for approximation issues. This technique has been extensively described in several scholarly sources (Li et al., 2023; Huang et al., 2023). The stochastic HTSWP coordination problem, which integrates CES systems, is expressed as follows:

$$\lambda = f(\varpi_p). \quad (26)$$

The symbols λ , f , and ϖ_p in Eq. (26) represent the output of the coordination issue involving CES systems in the HTSWP, the equations pertinent to the HTSWP, and the input parameter, respectively. In situations devoid of uncertainty, there is possibility for the anticipated conclusion to diverge from the observed findings. The energy produced by CESs may deviate from the anticipated values because to the inherent uncertainty linked to CESs' power output. When considering m random variables, Eq. (26) may be restated as follows:

$$\lambda = f(c, n_1, n_2, \dots, n_m). \quad (27)$$

In Eq. (27), the symbols c , n_y (where $y = 1, 2, \dots, m$), and Df_{ny} denote specific deterministic input variables, a set of random input variables, and the density function, respectively. Modelling input random variables (IRVs) statistically requires undivided attention to their beginning central moments. To calculate these moments, the following expression is employed:

$$\begin{aligned} \lambda_{(y,rs)} &= f(c, \mu_{n1}, \mu_{n2}, \dots, \mu_{y,rs}, \dots, \mu_{nm}), \\ rs &= 1, 2; y = 1, 2, \dots, m. \end{aligned} \quad (28)$$

Within expression (28), the variables n_{y1} and n_{y2} denote specific IRV. The mean and standard deviation are derived as solutions of $\lambda_{(y,rs)}$.

- **Step 1:** Establish a value for m corresponding to each of the IRV.
- **Step 2:** Specify that the expected value of λ^u is equal to zero, where u can take on values of 1 or 2.
- **Step 3:** Select an uncertain variable (UV) denoted as n_y .

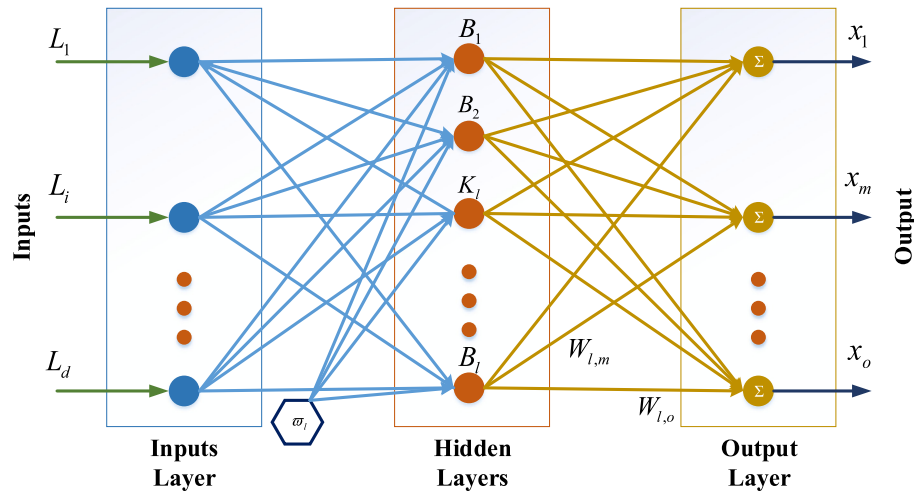


Fig. 1. Multiple-input-output radial basis function neural networks.

- **Step 4:** Calculate the skewness of the UV employing the subsequent formula.

$$\varphi_{ny,3} = E \left[(n_y - \mu_{ny})^3 \right] / \sigma_{ny}^3. \quad (29)$$

- **Step 5:** Utilize the provided expression to obtain stats with precision up to two decimal places.

$$\Gamma = \frac{\varphi_{ny,3}}{2} + (i^2)^{3-rs} \sqrt{\left(m + \left(\frac{\varphi_{ny,3}}{2} \right)^2 \right)}. \quad (30)$$

- **Step 6:** Calculate two potential positions.

$$n_{y,p} = \mu_{ny} + \Gamma_{ny,p} \cdot \sigma_{ny}, \quad rs = 1, 2. \quad (31)$$

- **Step 7:** Determine two potential parameter values in a deterministic manner for the coordination of all power sources.

$$\lambda_{(y,rs)} = f(\mu_{n1}, \mu_{n2}, \dots, n_y, rs, \dots, \mu_{nm}). \quad (32)$$

- **Step 8:** Next, calculate the weights parameter.

$$R_{(y,rs)} = \frac{(i^2)^{rs}}{m} \frac{\Gamma_{e,3} - rs}{\Gamma_{e,1} - \Gamma_{e,2}}, \quad rs = 1, 2. \quad (33)$$

- **Step 9:** Raise the moments by plugging them into the following procedure.

$$E(\lambda^u) = E(\lambda^u) + \sum_{rs=1}^2 R_{(y,rs)} \cdot [\lambda(y,rs)]^u. \quad (34)$$

- **Step 10:** Iterate through steps 4 to 9 until the UV has been covered.
- **Step 11:** Determine the subsequent.

$$\mu = E(\lambda^1), \quad \sigma = \sqrt{E(\lambda^2) - (E(\lambda^1))^2}. \quad (35)$$

The use of Weibull and Beta distribution functions in CESs data necessitates the determination of precise value of parameters that most accurately depict the data being collected. Point estimation seeks to offer a singular, optimal approximation for a parameter of interest, reducing the distribution to a precise point inside the area of the parameter space. In the case of wind power, the point estimate for the Weibull distribution entails finding the shape parameter as well as scale parameter by matching the gathered wind velocity information to the range. The point estimates obtained from this procedure provide a succinct depiction of the wind resource's attributes, such as the velocity of the wind with the highest likelihood. Similarly, in the context of solar power uses, the process of point estimating involves using the

Beta distribution to determine the shape parameters by matching solar irradiance statistics. These characteristics aid in defining the range of variation and likelihood distribution of solar energy.

Unlike thermal and hydroelectric plants that can generate energy regardless of weather conditions, CES relies heavily on meteorological factors to produce power. Therefore, it becomes crucial to establish an effective mechanism for handling the unpredictability of meteorological variations. In this study, the uncertain characteristics of CES are simulated using the (9) and the (15). To create a more accurate and realistic energy generation model, the research leverages climate data from the RETScreen software, sourced from global satellites maintained by the National Aeronautics and Space Administration (NASA). For comprehensive details regarding CES climate data and the specific plant site, please refer to Section 5.

4. HTSWP computing paradigm

The current study implements an optimization paradigm that combines the black widow optimization (BWO) algorithm with the radial basis function neural network (RBFNN). The hybrid approach has been utilized in numerous instances of restricted optimization to assess its effectiveness and efficiency. The RBFNN is categorized as an artificial neural network (ANN) and employs activation functions that are derived from Gaussian kernels. As per the research conducted in Yang et al. (2022), this model's architectural design possesses several benefits. The advantages encompass a straightforward and instinctive layout, the capacity to effortlessly modify parameters, and a notable level of generalization skills. The enhancement of ANN performance relies on the selection of suitable parameters. Two main methods commonly used for optimizing artificial neural network (ANN) variables are gradient-based techniques and metaheuristic computation. The previous method, utilizing techniques like BWO as illustrations, has a greater probability of avoiding local optima. The incorporation of BWO algorithms with ANNs has been extensively studied, encompassing many optimization issues (Xu and Wang, 2023).

4.1. RBF neural network (RBFNN)

The conducted experiment involved a three-tiered radial basis function neural network (RBFNN) comprising of an input layer, a hidden layer, and an output layer. The intricate arrangement of this network architecture is visually illustrated in Fig. 1. An activation method is used to build the connection between the input layer and the hidden layer. The activation function used in the hidden layer incorporates a Gaussian kernel function, denoted by Eq. (36) in mathematical notation (see Fig. 4).

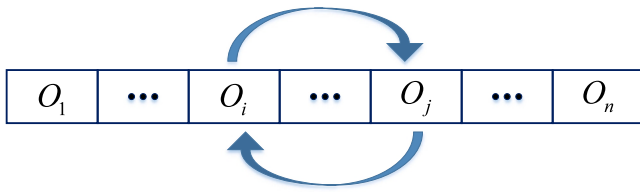


Fig. 2. The phenomenon of mutation in the actions of a single spider.

$$B_l = \frac{e^{-\|L - K_l\|^2}}{2\varpi_l^2} \quad (36)$$

$$x_m = \sum_{l=1}^B W_{l,m} B_l \quad (37)$$

RBF design is a specific type of multilayer feedforward neural network (MLFNN) that frequently uses a sigmoidal form for the activation functions of hidden neurons in MLFNN models, which is widely accepted in the field of machine learning. In contrast, the study (Al-Betar et al., 2023) highlighted that RBF networks commonly employ Gaussian basis functions as their primary approach. RBF Neural Networks (RBFNNs) employ a Gaussian distribution-based activation function for each neuron in the hidden layer. This function assesses the identical attribute among various units, hence enhancing the network's ability to detect patterns. The Gaussian basis function's exceptional capacity to accurately map individual data points to a feature space with an infinite number of dimensions provides a valid rationale for favouring it over alternative activation functions. Mapping is a procedure that transforms data that cannot be separated in a straight line into a format that can be separated in a straight line. The Eq. (37) represents the mathematical connection between the hidden layer and the output layer.

In this context, d , B , and o represent the number of neurons in the input layer, the hidden layer, and the output layer, respectively. The variables L , B_l , and x_m represent the individual neurons located in the following positions: The symbol L denotes the i th neuron in the input layer, B_l represents the l th neuron in the hidden layer, and x_m symbolizes the m th neuron in the output layer. The input is denoted as a vector x , which is composed of elements $L = [L_1, L_2, \dots, L_d]^T$. The weight $W_{L,m}$ represents the connection strength between the l th neuron in the hidden layer and the m th neuron in the output layer. The centre vector K_l corresponds to the l th neuron in the hidden layer, and ϖ_l represents the width of this neuron. The phrase $\|L - K_l\|$ denotes the Euclidean distance between the input vector and the centre vector. The overall scale of the RBFNN is determined by the number of neurons in the hidden layer, which determines its topological dimension.

4.2. Black widow optimization approach for training RBFNNs

Swarm intelligence algorithm falls under the category of meta-heuristic framework, which are well-established optimization methods known for their robust theoretical framework and impressive optimization capabilities. Like its counterparts in the swarm intelligence family, the black widow optimization approach (BWOA), as presented by Hayyolalam and Kazem (2020), pursues optimal solutions by emulating the reproductive behaviour of black widow spiders. This behaviour includes actions such as crossover, cannibalism, and mutation. The BWOA is a nature-inspired optimization algorithm that emulates the hunting habits of black widow spiders in order to tackle intricate optimization tasks such as HTSWP. During the web building stage, prospective solutions, symbolized as spiders, form webs within the search region. These webs encompass various areas of the solution area, resulting in a varied population of potential options. The hunting cycle emulates the predatory conduct of spiders as they navigate

and seek out potential prey. Within the method's framework, this stage entails the solutions systematically exploring the solution area, in search of the best possible outcomes or enhancements to the existing population. The technique employs a system that updates the solutions by evaluating their fitness using an objective function. Algorithmic processes undergo adjustments or updates in reaction to their effectiveness, hence providing an evolutionary element. Furthermore, the connection between structures and prey is of utmost importance, as it represents the exchange of information and cooperation across different solutions. This communication may involve the exchange of knowledge or the synchronization of motions in order to collectively enhance the total effectiveness of searches. The settings for parameters in BWOA consist of the group size (N), which determines the amount of spiders during every iteration, as well as the highest possible number of iterations ($MaxIter$), which specifies the whole execution time of the method. Web building characteristics dictate the formation of spider webs, impacting the variety and extent of the possible configurations. The hunting parameters regulate the process of exploration as well as exploitation in spiders, determining their movement and search strategies for finding the best solutions. The fitness value is a crucial parameter that determines the goal that the strategy aims to optimize. Like other optimizations techniques, it is crucial to carefully adjust these variables, as their ideal values can differ based on the particular optimizations problem being addressed. The BWOA is characterized by its swift convergence rate and high level of optimization accuracy.

The operational steps of the BWOA can be outlined as follows:

- **Step 1.** The BWOA commences by creating a population of random spiders. The step represents each spider entity through a possible solution vector.
- **Step 2.** The fitness function of the problem is specifically designed to assess the effectiveness of the solution. When robust offspring thrive, we see the maintenance of alternative solutions that have lower fitness values for the function. The most favourable solution is kept, while the less favourable one is discarded.
- **Step 3.** Crossover action is the mechanism by which both parent entities generate offspring, and this process is influenced by the crossover rate. The mathematical representation of this crossover behaviour is as follows:

$$\begin{cases} S_1 = \delta \times O_1 + (1 - \delta) \times O_2, \\ S_2 = \delta \times O_2 + (1 - \delta) \times O_1. \end{cases} \quad (38)$$

Each entry in the array ' δ ' is a randomly generated value that falls within the range of 0 to 1. The entities O_1 and O_2 serve as the parents, and the offspring are denoted as S_1 and S_2 . The crossover step is repeated $dim/2$ times. Cannibalism refers to the occurrence in which powerful female spiders consume weaker male spiders within the parental group, as well as when strong individuals consume weaker ones among the young. Mutation behaviour pertains to the mechanism by which a primary organism produces a subsequent organism, governed by the frequency at which mutations take place. The mutation process is depicted in Fig. 2, wherein elements such as $[O_1, \dots, O_i, \dots, O_j, \dots, O_n]$ undergo a transformation to $[O_1, \dots, O_j, \dots, O_i, \dots, O_n]$. The purpose of this stage is to investigate additional persons inside the search space, increase the diversity within the population, and update the potential solutions.

- **Step 4.** Iterate between Step 2 and Step 3 in a cyclical manner. Keep refining candidate solutions and evaluating their fitness function values persistently until either the predefined maximum iteration count is achieved or convergence towards the optimal value is attained.

Despite its impressive efficacy, the BWOA carries the risk of prematurely converging towards a specific point. To avoid this issue, the $\odot$ array is no longer regarded as a random variable. Instead, it undergoes

Table 2
Pseudo code of Proposed computing paradigm.

IBWOA-RBFNN Pesudo code for HTSWP
1-Initialize population size, generations, and algorithm hyperparameters
2-Initialize parameters for IBWOA
3-Initialize RBFNN architecture, learning rate, and training parameters
4-Initialize a population of diverse black widow agents with varying traits
5-Create an initial set of RBFNN weights and biases
6-Define an objective function to evaluate the fitness of HTSWP scheduling solutions
7-For each generation in the range(generations):
8-For each black widow agent in the population:
9-Apply exploratory and exploitative movements using predefined rules
10-Evaluate fitness by integrating the RBFNN with the HTSWP model
11-If the obtained fitness is superior to the previous best:
12-Update the best solution and corresponding RBFNN parameters
13-Perform selective mating and selection based on fitness scores
14-Apply advanced crossover and mutation operators to produce diverse offspring
15-Update the population with the newly generated offspring
End For
Return the optimal HTSWP solution
End IBWOA-RBFNN

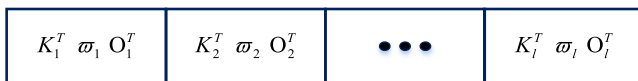


Fig. 3. Black widow's unique anatomy.

a nonlinear evolution over time. The equation that governs the $\mathbb{O}$ array in the enhanced version of the (BWOA) is expressed as follows:

$$\mathbb{O}_i = 8 \times (1.23/(\text{iter})^{0.3}) \quad \mathbb{O} = [\mathbb{O}_1, \mathbb{O}_2, \dots, \mathbb{O}_n] \quad (39)$$

Hence, the formula for updating the crossover procedure is as follows:

$$\begin{cases} S_1 = \delta \times O_1 + (1 - \delta) \times O_2, \\ S_2 = \mathbb{O} \times O_2 + (1 - \mathbb{O}) \times O_1. \end{cases} \quad (40)$$

4.3. The structural framework of the IBWO-RBFNNs design

The IBWOA collectively investigates the centre vector, width, and output vector in the context of the RBFNN. The connection between the IBWOA and the RBFNN is formed by applying potential solution vectors derived from the BWOA, as shown in Fig. 3.

The dimensions of the IBWOA-RBFNN framework are dictated by the magnitude of the hidden layer neurons. Thus, the symbol 'h' is employed to denote both the quantity of hidden layer neurons and the overall magnitude of the neural network. The proportions of the black widow are determined by quantifying the amount of neurons in the input, hidden, and output layers of the IBWOA-RBFNN architecture. The variable D represents the magnitude of the black widow, whereas the variable d signifies the quantity of input vectors, and the variable o represents the quantity of output vectors. The number of neurons in the input layer of the IBWOA-RBFNN is dictated by the size of the input data. The architecture described here functions as a univariate model for regression tasks, but has the capability to transform into a multivariate model for limited optimization problems. An inherent constraint of combining IBWOA with RBFNN is its reliance on global information to achieve the most favourable outcome, rendering it more susceptible to cyber attacks. Moreover, in certain multi-dimensional issues, they require greater processing resources, such as time and memory, in comparison to standalone metaheuristic techniques. This restricts their usefulness in contexts with limited resources. The determination of the number of neurons in the output layer of the IBWOA-RBFNN is contingent upon the categories present in optimization tasks.

The performance of the IBWOA-RBFNN is affected by essential variables such as population density and iteration frequency. The selection of reproductive factors, such as the degree of crossover as

well as mutation, directly influences the level of diversity within the solution space. The selection process and predefined search space are crucial factors that significantly influence the algorithm's behaviour. Optimizing these parameters is essential for attaining efficient results in a wide range of problem domains (see Table 2), and the flowchat of proposed IBWO-RBFNN framework for the HTSWP is depicted in Fig. 4.

5. RETScreen (wind and solar dataset details)

RETScreen, also known as the RETScreen Clean Energy Project Analysis Software, is a dynamic tool built on Microsoft Excel that uses complex computational models to aid in the discovery of renewable energy sources. RETScreen was developed at Canada's Canmet Energy Research Center in 1996 and has since received widespread acclaim for its ability to perform complex analyses on a wide range of variables, such as those related to green energy financing, risk assessment, and greenhouse gas emissions. Education on renewable energy, capacity building, and decision-making are just some of the many uses that might be made of this innovative and unique instrument. This program is useful for determining the feasibility of renewable energy projects. Its capacity to function with a smaller data footprint and fewer computing resources compared to other techniques is only one of its many benefits (Akpahou et al., 2023).

This study employed the RETScreen programme for applying real-time climatic data for wind and solar resources in order to obtain HTSWP. The optimization of our proposed scheduling model, which relies on CESSs, becomes considerably more intricate when include power loss and the valve point effect. In addition, the PET approach was utilized to specifically tackle the inherent uncertainty associated with CESSs, leading to a pragmatic and efficient solution for HTSWP. The climatic data diligently collected by NASA satellites is visually depicted in Fig. 5. The RETScreen Climate databases contains the weather information for the production of solar energy and the wind that is required for the initial development of the modelling. Weather information gathered from 6,700+ local monitoring centres, as well as statistical information from NASA's international satellites, and is accessible through the programme. Depending on the underlying data sources and modelling techniques used, RETScreen's solar and wind data is accurate and reliable. To model the efficiency of CESSs, the programme consults a number of meteorological databases, including NASA's Surface Meteorology and Solar Energy (SSE) library. To accurately assess the performance of solar energy systems, good simulations require correct solar radiation data as in the proposed approach PET is employed, which statistically adjust the inherent uncertainties. The reliability of RETScreen's findings is heavily dependent on the modelling techniques it employs. Such algorithms must be validated and calibrated with

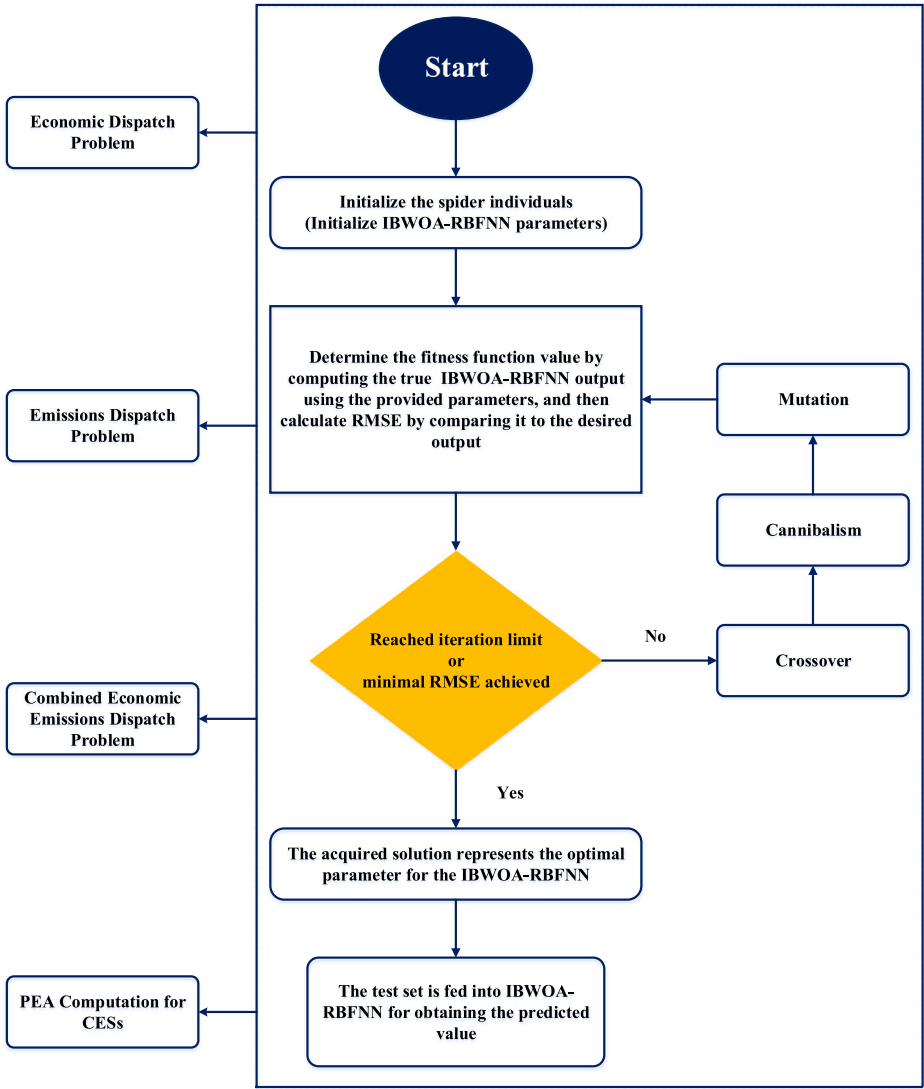


Fig. 4. The framework for the IBWO-RBFNN upgraded black widow optimization for HTSWP.

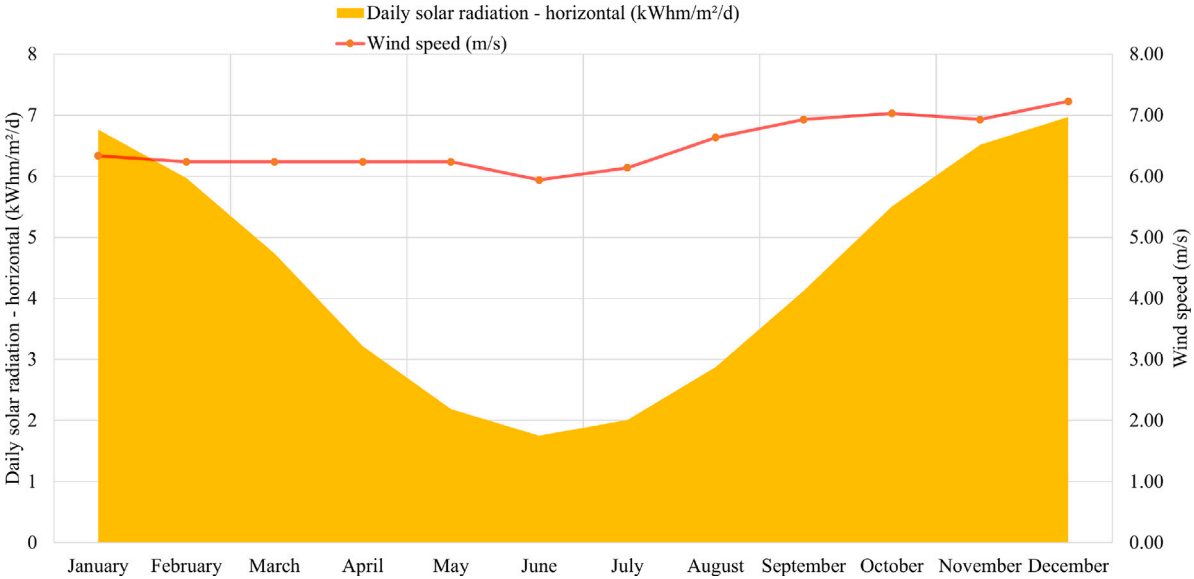


Fig. 5. CESs data statics computed by RETScreen (wind and solar).

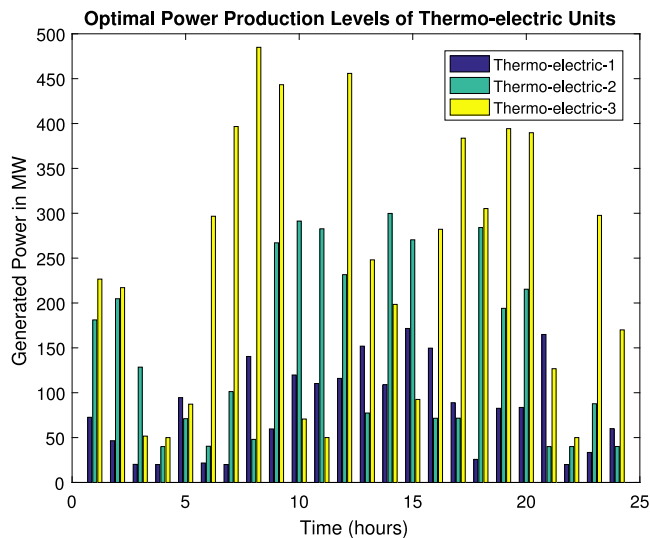


Fig. 6. Thermo-electric generation levels computed by IBWOA-RBFNN.

real-world data from different places and climates to guarantee their trustworthiness. Users can acquire the NASA Near Real-time Global Radiation and Meteorology statistics set through the “Acquire NASA dataset” option in the RETScreen software. This set allows users to obtain historical and near real-time solar and atmospheric information on a daily basis (Agyemang-Boakye et al., 2023).

6. Results and simulation

The dependability of the system that has been implemented, which combines a scheduling framework with CESs, has been evaluated using the IBWOA-RBFNN approach. The IBWOA and RBFNNs combine to generate a powerful optimization synergy. Nature-inspired metaheuristics help the IBWOA explore solution areas globally. Integrating this with RBFNNs leverages the network’s local adaptation and fine-tuning. This combination optimizes the algorithm’s global search and the neural network’s convergence precision, improving efficiency. The hybrid metaheuristic accelerates convergence and performs well on many complex optimization problems. The IBWO-RBFNN stands out due to its unique features and the combined advantages it offers compared to other optimization approaches (Dasgupta et al., 2022; Zhang et al., 2017; Fang et al., 2014; Lakshminarasimman and Subramanian, 2006). IBWO, known for its exceptional ability in global optimization, excels in thoroughly investigating extensive solution spaces and achieving a careful equilibrium among exploration as well as exploitation. The effective convergence mechanism of this method allows a rapid and comprehensive exploration of the search space, leading to speedier convergence when compared to other methods. The versatility and durability of IBWO further boost its appeal, making it appropriate for a wide range of optimization tasks such as HTSWP. The incorporation of IBWO-RBFNN uses the advantages of both methodologies. The RBFNN, known for its ability to recognize patterns and generalize, serves as a valuable addition to the global optimization strategy of IBWO. The integrated strategy utilizes both learning and adjustment, allowing the procedure to make well-informed decisions throughout the optimization process. This integration is not only logically sound but also empirically confirmed through the presented case studies. The suggested method’s outstanding effectiveness with regard to rapid convergence, high solution quality, and general resilience confirms its

effectiveness, strengthening the case for its favour over alternative optimization strategies. It is important to acknowledge that the efficiency of this integrated strategy may differ depending on the specific characteristics of the optimization challenge. The present assessment encompassed the examination of several test situations, each of which was accompanied by unique system limitations. The system under consideration comprises of a total of four hydro-electric units that are interconnected, along with three thermal-electric units. Additionally, there are two CESs incorporated in the system, specifically wind generation and solar power facilities. All of the system’s energy resources have had their initial information as well as parameters retrieved from the Dasgupta et al. (2022). In our study, each solar system has thirteen modules with different energy production costs. Maximum output capacity, energy production costs, and operational limits impact photovoltaic system performance. This work tested numerous situations to evaluate our systems. IBWOA-RBFNN explored objective functions relating to production cost, emissions, and a combination of both in these situations. Our goal was to reduce scheduling model functions. This section discusses using CESs in planning tasks and shows how the suggested IBWOA-RBFNN algorithm may solve classic and novel scheduling challenges.

6.1. Case study-I (Economic dispatch analysis)

This test system’s objective is to assess the efficacy of the integrated economic dispatch task (EDT) when coupled with CESs such as photovoltaic and wind systems. The incorporation of CESs into the conventional EDT model presents a scheduling challenge. The main goal is to minimize power generation expenses without compromising efficiency, all while dealing with the system-associated limitations of the traditional integrated EDT model with CESs. The cost of producing energy from all sources is determined using Eq. (18) as the framework’s target metric. The IBWOA-RBFNN method has been effectively applied to both the traditional EDT model and the proposed scheduling complexities associated with the photovoltaic-wind-based system in order to determine the lowest practicable production costs. Achieving cost-effective energy production involves optimizing the utilization of all available production units while minimizing operational expenses. To achieve this objective, the IBWOA-RBFNN method is utilized to oversee the hourly water discharge from hydro-electric units and the energy production from thermal power plants. The power patterns achieved by these facilities, in conjunction with the water discharge throughout a 24 h duration, are visually presented in Figs. 6 and 7, respectively.

In this particular case study, the simulation results unequivocally establish the superior performance of the IBWOA-RBFNN approach compared to other advanced computational methods in addressing the HTSWP. This superiority is underscored by a direct comparison with recent studies in the literature, as cited in Dasgupta et al. (2022). Notably, our proposed model distinguishes itself by its heightened practicality and realism when contrasted with studies referenced in Dasgupta et al. (2022), Zhang et al. (2017), Fang et al. (2014), Lakshminarasimman and Subramanian (2006). The increased practicality stems from the incorporation of a more extensive array of real-world constraints, elucidated in Eqs. (19), (20), (21), (22), and (23). The relevant costs associated with all power resources, emissions levels from thermal plants, and the simulation time required for optimization are consistently lower when compared to alternative advanced approaches. In the conventional scheduling problem context, IBWOA-RBFNN attains the best global optimum operational cost, totalling 3.6e4 \$/day, surpassing results found in Dasgupta et al. (2022). Additionally, IBWOA-RBFNN excels in calculating emissions from thermo-electric power plants, registering at 2.8e4 lb/day, with line losses measured at 40.7 MW. The convergence behaviour of IBWOA-RBFNN, illustrated in Fig. 8, further accentuates its remarkable capability to rapidly reach the global optimum solution within a mere 22 iterations in 14.1 s, proficiently addressing the systemic constraints inherent in HTSWP scheduling.

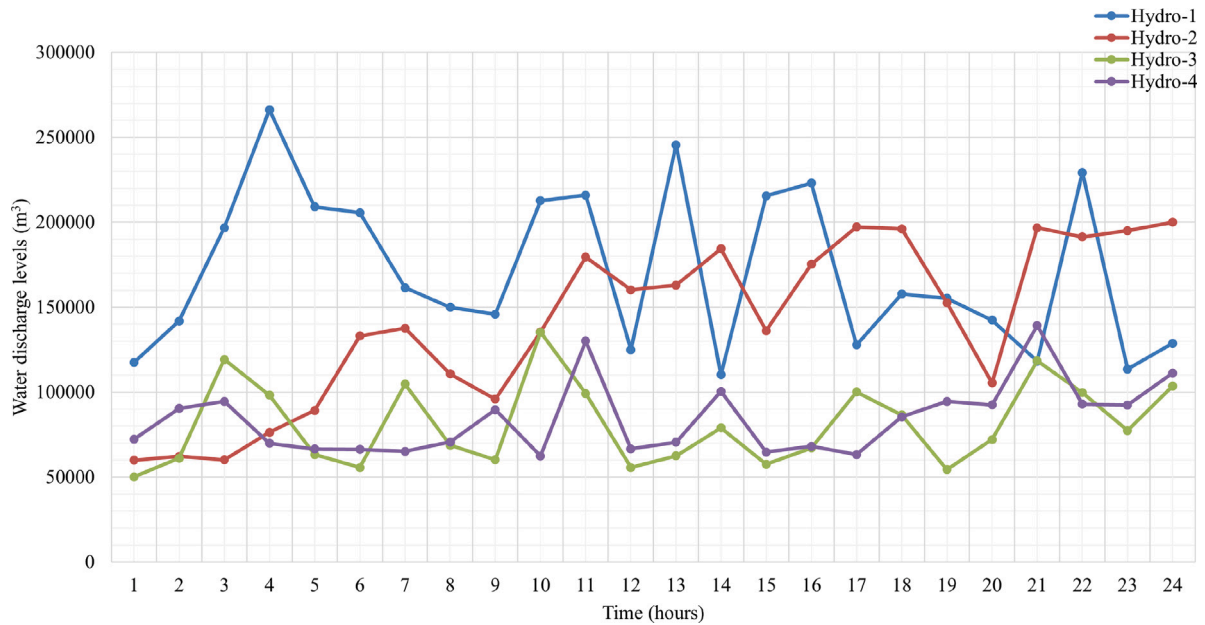


Fig. 7. Water discharge levels of hydro-plants.

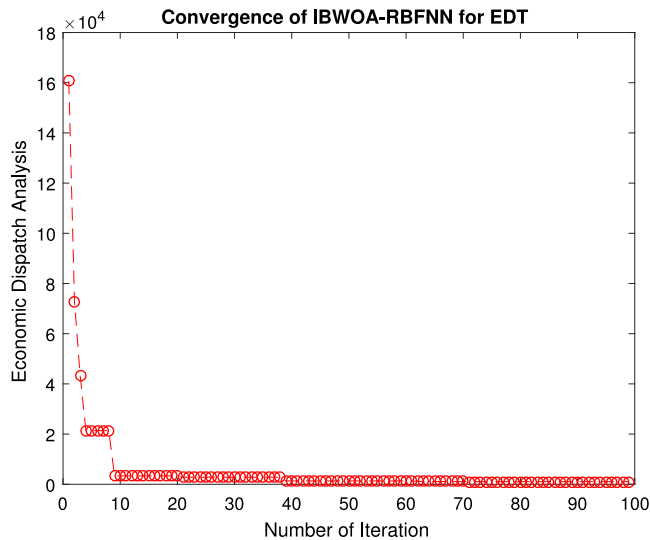


Fig. 8. IBWOA-RBFNN convergence for Case Study-I.

The majority of thermo-electric units rely on fossil fuels, a practice associated with both harmful post-combustion emissions and significant economic costs due to the annual expenditure on imported fuel running into millions of dollars. To address this challenge, enhancing the reliability and efficiency of the energy production process is essential. One effective approach involves augmenting the power supply with green sources like photovoltaic modules and wind generators.

The integration of CESs into the energy generation mix presents numerous benefits. It enables energy managers and partners to maximize the cost-efficiency of carbon-neutral energy generation while simultaneously decreasing polluting emissions through decreased reliance on imported fuels. Additionally, it provides a means for power generation companies to contribute to the fight against global warming and incentivizes entrepreneurs to invest in renewable energy production

equipment. The adoption of the IBWOA-RBFNN approach plays a crucial role in maximizing energy output. This is achieved by considering practical factors related to wind and photovoltaic systems, such as solar irradiance and temperature coefficients, thereby ensuring the efficient utilization of these renewable resources.

The effectiveness of our proposed approach by analysing the time periods during which production costs are minimized. In hydroelectric production, two critical factors are the speed of water outflow and the reserve level of water in the facility. Additionally, the IBWOA-RBFNN algorithm is utilized to calculate the water flow benefits for hydroelectric power and reduce the operational costs associated with thermoelectric, wind, and photovoltaic energy generation. The results of this computation are displayed in Figs. 9, 10, and 11. The convergence profile of the proposed IBWOA-RBFNN including CESs for Case Study-I is shown in Fig. 12. These figures illustrate the amount of water discharged through every reservoir, as well as the energy production from hydroelectric, thermoelectric, solar power, and wind facilities. These values were determined using the IBWOA-RBFNN method.

Comparatively, in contrast to a recent study mentioned in Dasgupta et al. (2022), our results indicate that the computed fuel cost, emissions, and power losses are 2.6e4 \$/day, 7.7e4 lb/day, and 399.129 MW, respectively. Additionally, the production costs for photovoltaic and wind energy are 1148 \$/day and 3055 \$/day, respectively.

6.2. Case study-II (emissions dispatch analysis)

The primary objective of the scheduling operation in this situation is to improve the efficiency of generating electricity and lower the emission levels within a 24 h period. Conventional energy production from thermo-electric sources often relies on fossil fuels, which, in the electricity generation process, emit a range of contaminants into the environment, including as CO_x , NO_x , SO_x , and other substances. To address environmental sustainability concerns, various international institutions, including the United Nations, impose restrictions on the emissions generated by energy-producing entities.

In response to these challenges, a modern production framework has been developed by integrating traditional scheduling practices with

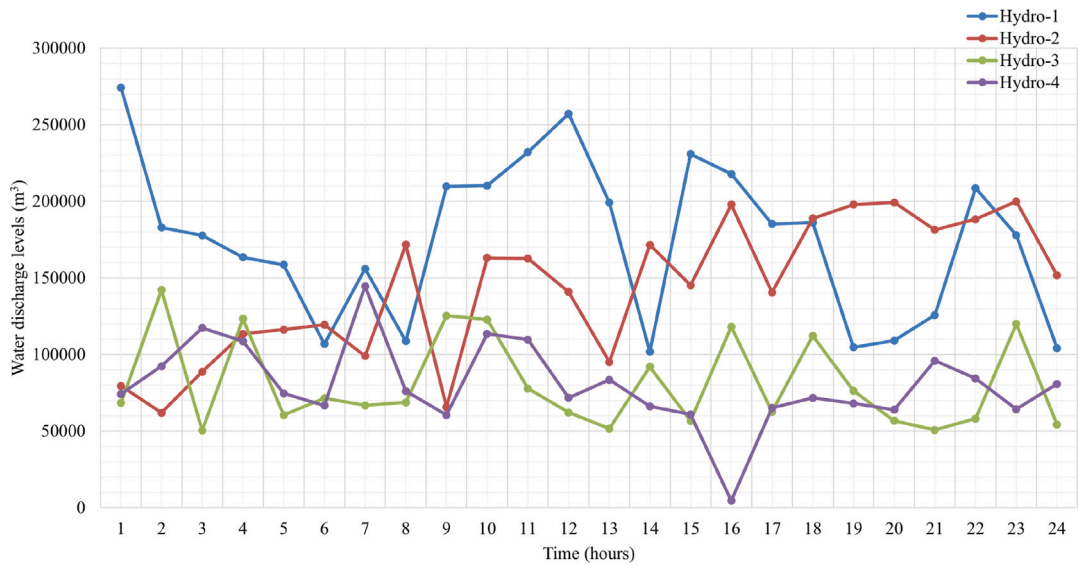


Fig. 9. Water discharge levels of hydro-plants.

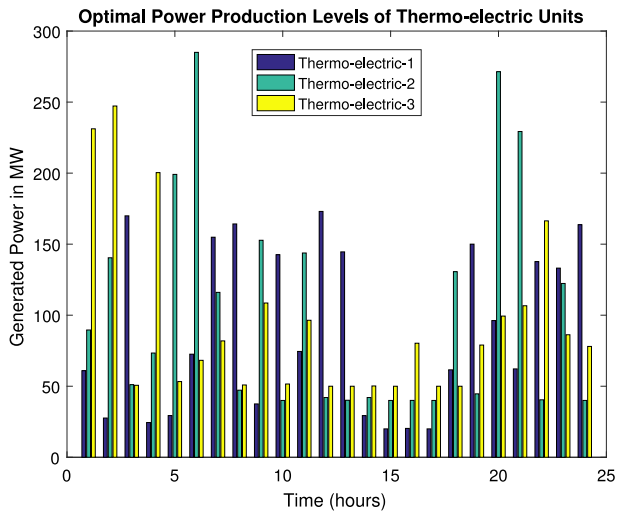


Fig. 10. Thermo-electric generation levels computed by IBWOA-RBFNN (including CESs).

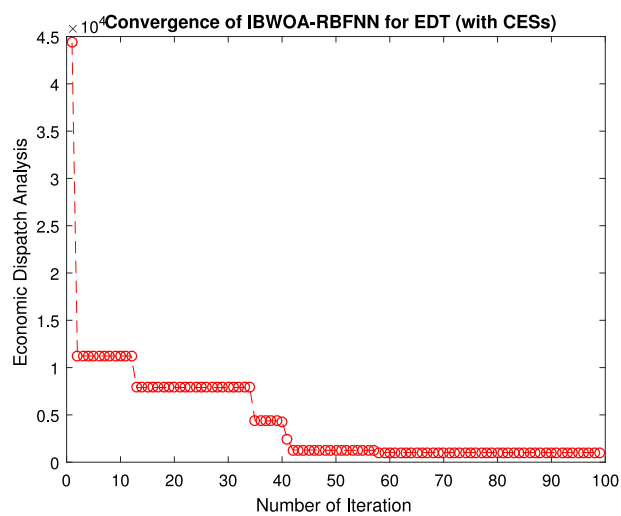


Fig. 12. IBWOA-RBFNN convergence for Case Study-I (including CESs).

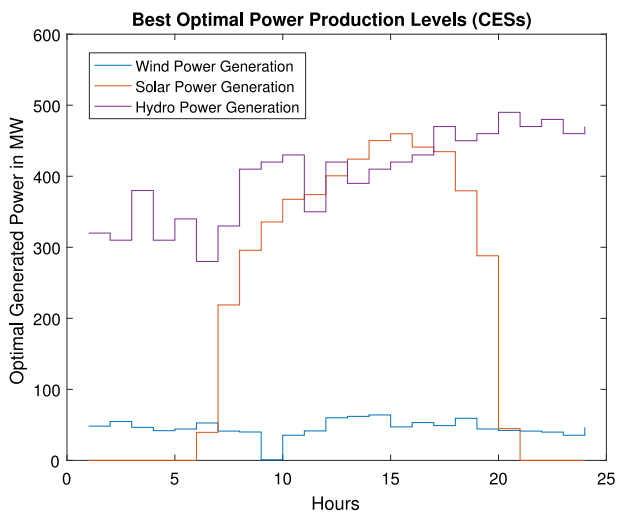


Fig. 11. Solar-wind and hydro powers levels computed by IBWOA-RBFNN.

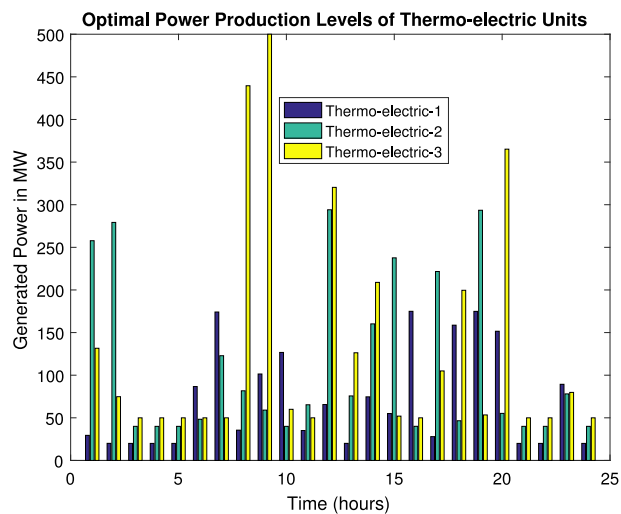


Fig. 13. Thermo-electric generation levels computed by IBWOA-RBFNN.

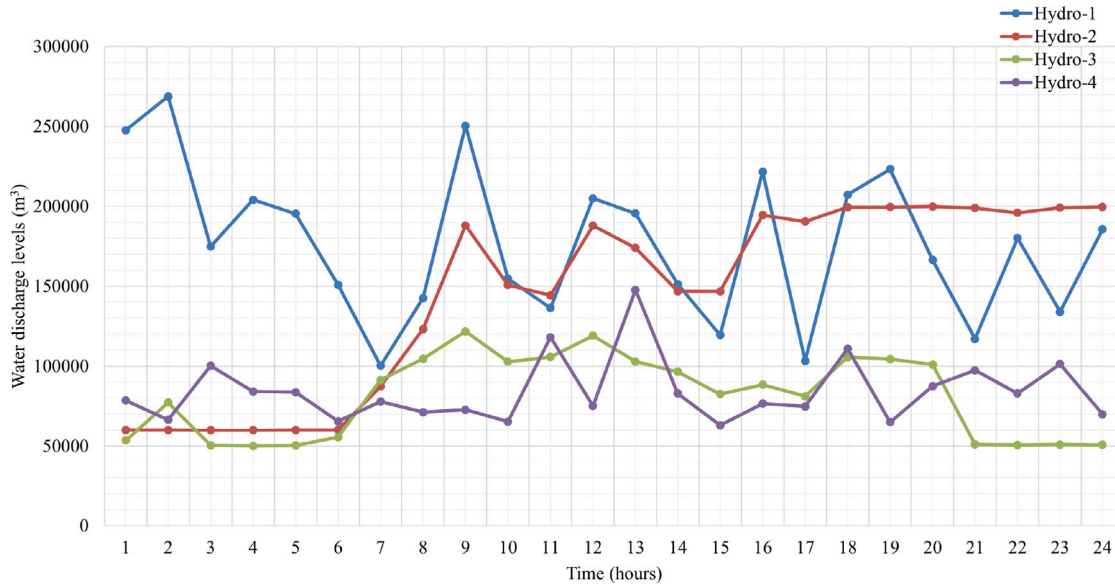


Fig. 14. Water discharge levels of hydro-plants (Case Study-II).

CESs such as photovoltaic and wind energy. The aim of this study was to create an efficient and environmentally friendly electricity production framework that ensures optimal emission levels. To evaluate the effectiveness of this framework, an emission dispatch analysis was conducted.

At first, our analysis concentrated on the EES described in Eq. (2) during this specific test scenario. This work utilized the γ IBWOA-RBFNN procedure to effectively minimize the emissions resulting from the combustion of fossil fuels in thermo-electric facilities. IBWOA-RBFNN achieves an optimal balance between emissions and power generation levels for the generators, ensuring that power balance constraints, as outlined in Eqs. (19)–(23), are full-filled. Figs. 13 and 14 provide visual representations of the calculated production unit power levels and water outflow rates over a span of 24 h.

In contrast with recent study in Dasgupta et al. (2022), for traditional emission dispatch analysis, IBWOA-RBFNN outperforms other advanced methods by achieving the best optimal balance between emissions and generation cost. Specifically, IBWOA-RBFNN attains emissions of 15887.42 lb/day and an energy production cost of 26072.89 \$/day for this scenario. Additionally, the computed power loss in this context is found to be 39.0337 MW. Furthermore, the obtained results clearly demonstrate that IBWOA-RBFNN outperforms other optimization methods such as the CSA algorithm, the WOA strategy, and the BSO algorithm. In comparison, IBWOA-RBFNN achieves a more substantial reduction in emissions, surpassing the alternatives by 828.17 lb/day, 1454.74 lb/day, and 1689 lb/day, respectively.

It is important to highlight, as evident in Fig. 15, that the proposed IBWOA-RBFNN method exhibits a swift convergence towards a feasible area of solution. This stands in stark contrast to the convergence patterns observed in advanced approaches as recently documented in Dasgupta et al. (2022). In this test, authors focused on addressing the emissions issue by analysing a production scheduling framework that relies on environmentally-friendly sources of electricity. Traditional scheduling methods were enhanced by incorporating CESs. In this study, an IBWOA-RBFNN algorithm was applied to establish an optimum generation coordination for the HTSWP with the primary objective of minimizing emissions. Fig. 8 depicts the data collected by IBWOA-RBFNN, showcasing the twenty four-hour water discharge rates from every reservoir as well as the energy production from all energy mix resources. A comprehensive comparative analysis of the

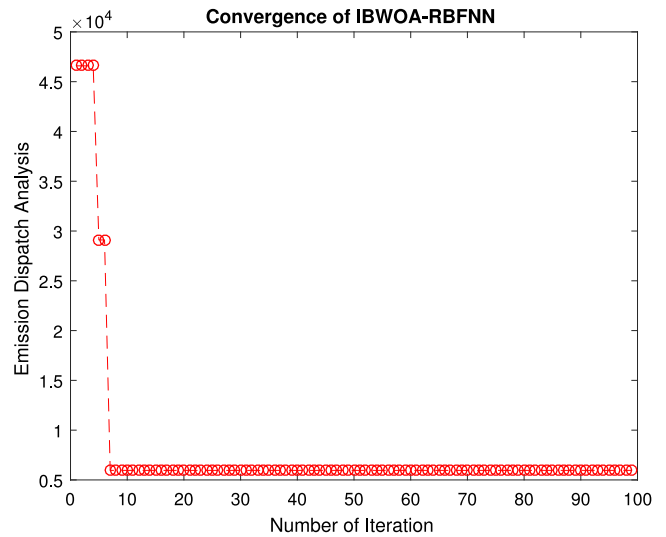


Fig. 15. IBWOA-RBFNN convergence for Case Study-II.

results in this specific case underscores the superior performance of IBWOA-RBFNN over other soft computing techniques in effectively managing an intricate coordination task with multiple goal functions. The overall operational expense of the scheduled HTSWP energy production amounted to 29,008.88 \$/day, while IBWOA-RBFNN achieved a total optimal emission of 5084.830 lb/day. Furthermore, the expenses associated with each individual resources, including thermal plants, photovoltaic generation facilities, and wind power plants, were 25,724 \$, 1238.6 \$, and 2045.5 \$, respectively. When compared with Dasgupta et al. (2022), the convergence pattern for this specific scenario is visually represented in Fig. 16, once again highlighting the remarkable capability of the proposed IBWOA-RBFNN method to swiftly converge towards securing a global optimal solution. In this scenario, the calculated power loss amounts to 389.1913 MW, which is notably 64

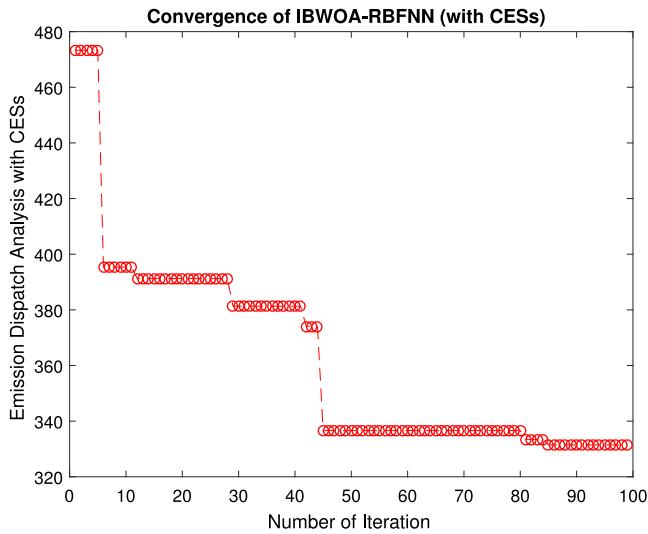


Fig. 16. IBWOA-RBFNN convergence for Case Study-II (including CESS).

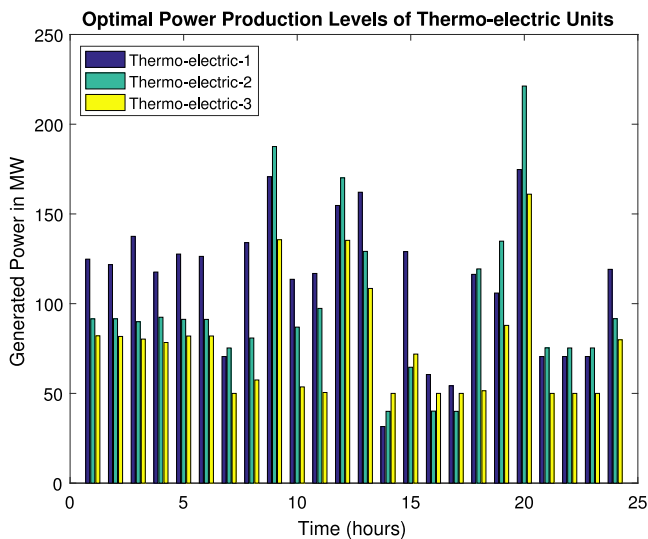


Fig. 17. Power levels of thermo-plants (Case study-II with CESS).

MW lower compared to the results obtained through the SCA approach (Dasgupta et al., 2022). The water discharge frequency in this specific scenario are visually presented in Fig. 18. Additionally, Figs. 17 and 19 depict the power production levels for thermals hubs and CESSs, respectively. These graphical representations highlight how the proposed IBWOA-RBFNN method effectively leveraged cost-effective generation sources and precise scheduling to address the complexities of a multi-objective emission function integrated with CESS.

6.3. Case study-III (Multi-objective economic emissions dispatch analysis)

This specific case, this study has conducted an analysis of both the manufacturing expenses and the environmental impact of fuel during the scheduling process. Authors of study utilized Eq. (3) as the objective function, which encompasses four hydro-electric and three thermo-electric facilities. To determine the value of “ h ,” this work calculated the ratio between Eqs. (1) and (2). Subsequently, the IBWOA-RBFNN algorithm was employed to minimize both functions F_{cost} and $F_{emission}$. By applying the IBWOA-RBFNN method appropriately, this work were

able to achieve the optimal value for $F_{composite}$, thereby determining the most suitable energy allocation for the scheduled hours.

Fig. 20 visually represents the water outflow levels that were computed using the IBWOA-RBFNN algorithm. Notably, the proposed approach has achieved an optimal cost of 26558 \$/day and emissions of 12932 lb/day, which, when compared to the results in Dasgupta et al. (2022), demonstrate a higher level of optimality. The energy production ranges for thermal plants resources are illustrated in Fig. 21, while the measured power losses during the span of twenty-four hours for this particular scenario amount to 37.7570 MW, IBWOA-RBFNN algorithm once again demonstrates superior performance compared to other approaches in this intricate setting, effectively reaching the optimal solution with high precision, as indicated in Fig. 22 for Eq. (3).

The comparative results clearly indicate that IBWOA-RBFNN outperforms other advanced approaches mentioned in Dasgupta et al. (2022), achieving optimal outcomes in every aspect. This study has extended the application of the developed production model to investigate a scenario that encompasses both economic and emission considerations concurrently. With the aim of minimizing energy production costs while reducing emissions during the span of twenty-four hours, wind power units and photovoltaic generation facilities were integrated into the conventional scheduling technique. Leveraging the IBWOA-RBFNN algorithm, we have successfully identified the most optimal time frames for electricity generation within our scheduled operations. Fig. 23 visually presents the water outflow levels calculated by IBWOA-RBFNN during the span of twenty-four hours, offering insights into the effectiveness of the proposed approach in achieving the dual objectives of cost reduction and emission minimization. In this scenario, once again, the proposed IBWOA-RBFNN method demonstrates excellence across various metrics, encompassing operational cost, emissions, and convergence performance. The convergence plot, depicted in Fig. 24, serves as a visual testament to the efficiency of the suggested optimizer, showcasing its swift convergence towards the global optimal solution for the specified objective function. This further underscores the robust performance of IBWOA-RBFNN in navigating the complexities of the scenario and achieving optimal outcomes.

In contrast to recently developed techniques outlined in Dasgupta et al. (2022), the IBWOA-RBFNN algorithm has yielded notable results in the examined scenario. It achieved an operational cost of 27,351.1958 \$/day, inclusive of contributions from CESSs, and emissions totalling 7379.746 lb/day. The cost associated with thermal power units for this specific case amounts to 23,054 \$/day, with the corresponding energy ranges illustrated in Fig. 25. Additionally, the expenses for photovoltaic power facilities is determined to be 1238.6 \$, while wind power generation costs 2045.50 \$. Fig. 26 visually presents the power levels achieved by the CESSs in this particular scenario.

The IBWOA-RBFNN algorithm achieves both optimal cost and emissions, while also accurately calculating power losses. Notably, the power losses obtained using this algorithm are significantly lower compared to those obtained using the CSA and WOA strategies. The comprehensive performance metrics of the proposed approach for all case studies are presented in Table 3, further highlighting the efficacy and superiority of IBWOA-RBFNN in contrast to other computing paradigms presented in the literature.

7. Conclusions and future research directions

The primary objective of this study was to demonstrate the effectiveness of the IBWOA-RBFNN method by presenting a practical scheduling framework for sustainable energy sources. Our research focused on analysing a hybrid energy hub scheduling problem, incorporating four hydroelectric, three thermoelectric, two wind, and two photovoltaic generation facilities. The developed framework stands out for its productivity and practicality, accounting for a more comprehensive set of system constraints compared to previous studies. IBWOA-RBFNN proves to be a more efficient, robust, and reliable

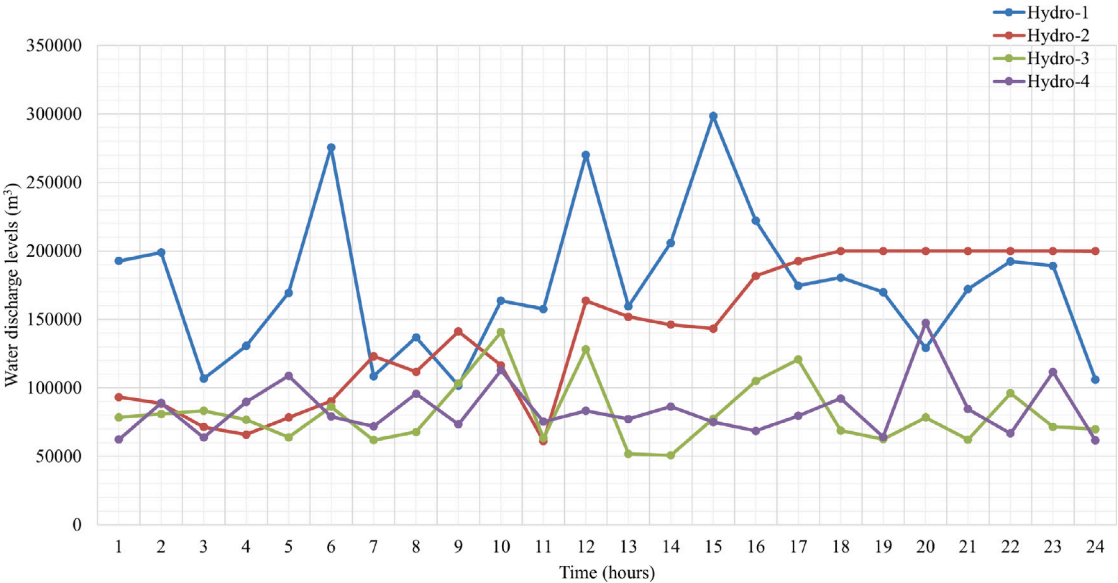


Fig. 18. Water discharge levels of hydro (Case Study-II with CESs).

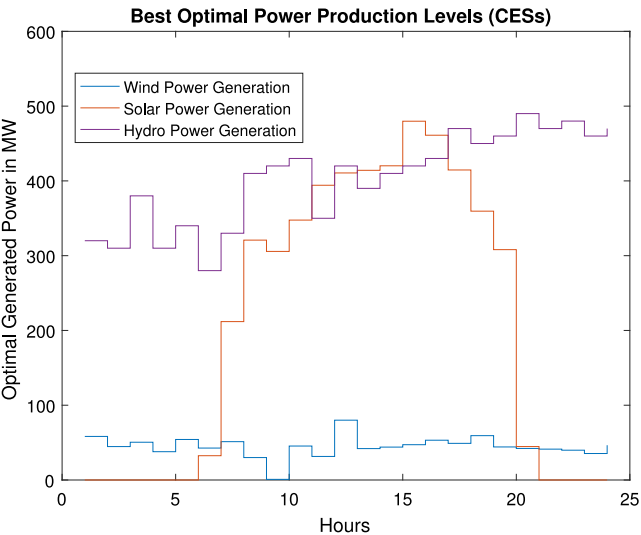


Fig. 19. Solar-wind and hydro powers levels computed by IBWOA-RBFNN (Case Study-II).

Table 3
Performance metrics of proposed IBWOA-RBFNNs.

Case study-I		
Max	Min	STD
1.6067e+05	1.0306e+03	1.8244e+04
Case study-II		
Max	Min	STD
4.6591e+04	5.9612e+03	8.5914e+03
Case study-III		
Max	Min	STD
1.5766e+05	1.4378e+03	3.2931e+04

tool, capable of reducing both the operational costs of electricity production and post-combustion emissions. Unlike previous works, our

framework benefits from more accurate and real data for Clean Energy Sources (CESs) obtained from RETScreen. Furthermore, IBWOA-RBFNN effectively utilizes its diverse control mechanisms to balance exploration and exploitation phases, leading to the identification of nearly optimal global solutions for specified objective functions in Case Study-I, II, and III. Moreover, when compared to competing techniques such as sine-cosine, back-search algorithm, whale optimization, small population-based particle swarm optimization, quantum-inspired evolutionary algorithms, and modified differential evolution, IBWOA-RBFNN emerges as the most effective and promising solution in terms of convergence, computational efficiency, and practical implementation. In Case Study-I, the best attained optimal cost and emissions are \$3.6e4/day and 2.8e4 lb/day, respectively, with photovoltaic and wind energy costs at \$1148/day and \$3055/day, respectively. For Case Study-II, a complex function, emissions of 15887.42 lb/day and an energy production cost of \$26072.89/day are achieved. For Case Study-III, the optimal cost of \$26558/day and emissions of 12932 lb/day are acquired using the proposed approach.

Although IBWOA-RBFNN is ideal tool for constrained optimization problems but integrating IBWOA with RBFNN has only one drawback: it needs global data to find the optimal response, making it more vulnerable to cyberattacks. They also require more computer resources, such as time and memory. Integrating IBWOA with RBFNN with distributed consensus-based methods is essential to make centralized optimization algorithms less vulnerable to cyberattacks. Attackers will have a hard time compromising the entire system since these techniques distributed control, use resilient cryptography, and use consensus mechanisms to distribute decision-making across numerous nodes.

To further evaluate the efficacy of the IBWOA-RBFNN algorithm in the HTSWP coordination model, it is possible to incorporate future innovations like hydrogen fuel cells, tribo-electric nano-generators, and centrifugal storage devices for energy storage. Research on economically viable mixed-energy dispatch algorithms is essential for the integration of these technologies. This will guarantee a continuous and cost-effective balance in the use of different energy storage methods, leading to more reliable and sustainable power systems. Furthermore, it would be valuable to investigate the possible uses of this technology in integrating electric vehicles (EVs), particularly in the context of vehicle-to-grid (V2G) operations. IBWOA-RBFNNs combines the adaptability of IBWOA with RBFNNs' nonlinear mapping, effectively capturing V2G

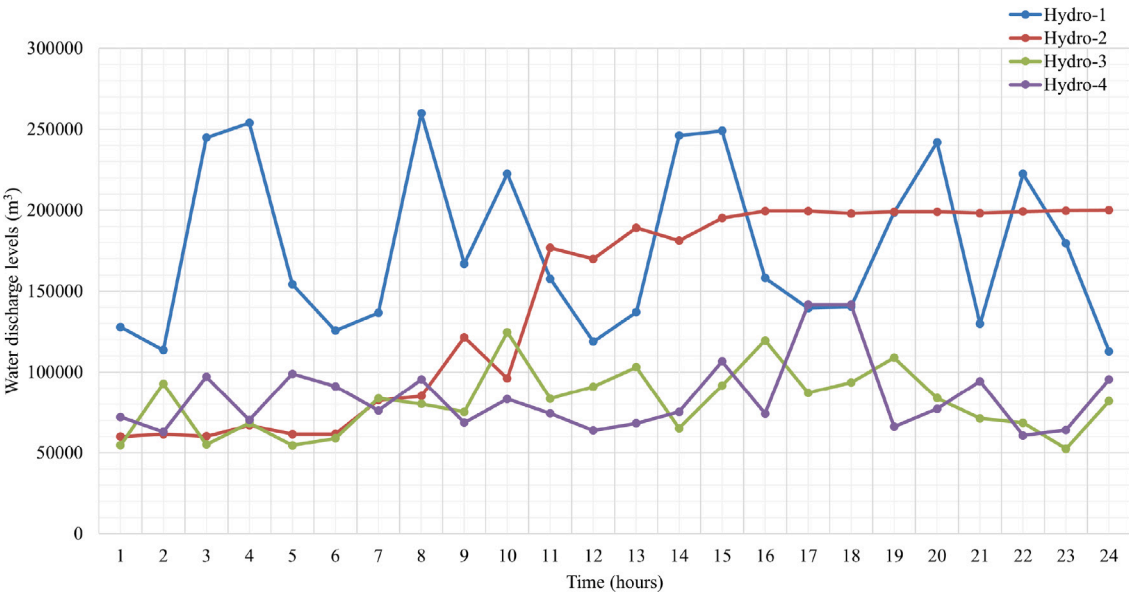


Fig. 20. Water discharge levels of hydro hubs (Case Study-III).

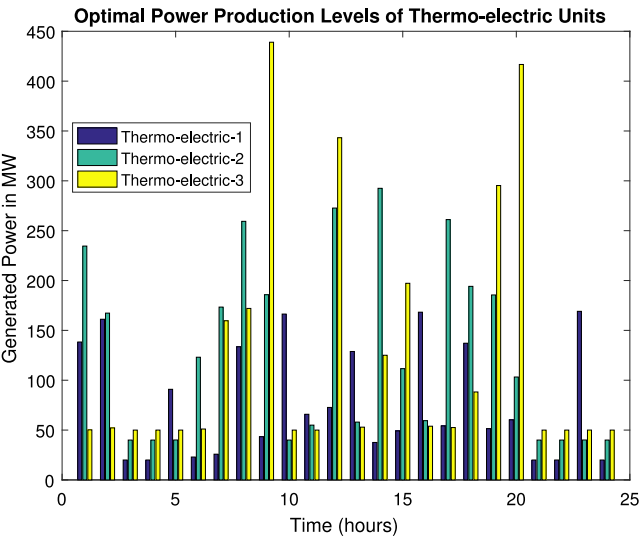


Fig. 21. Optimal power levels of thermal hubs (Case Study-III).

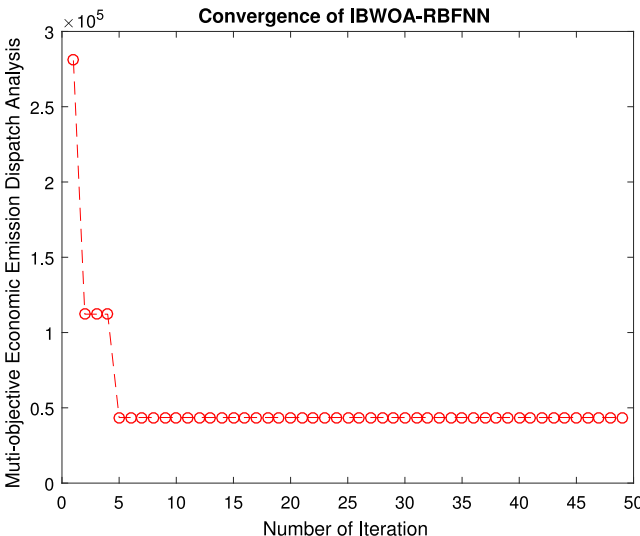


Fig. 22. IBWOA-RBFNN convergence for Case Study-III.

nonlinear dynamics. This hybrid approach optimizes by efficiently navigating diverse EVs recharging behaviours and grid load conditions, offering a robust framework for real-time decision-making. This enhances energy efficiency, grid stability, and overall V2G performance.

All authors confirms that they do not possess any identifiable conflicting financial interests or personal affiliations that might have influenced the research presented in this manuscript.

CRediT authorship contribution statement

Muhammad Khalid: Data curation, Methodology, Software, Writing – review & editing. **Ijaz Ahmed:** Conceptualization, Investigation, Supervision. **Mohammad AlMuhaini:** Formal analysis, Reviewing, Simulation. **Andrey V. Savkin:** Visualization, Resources, Project administration.

Declaration of competing interest

The authors declares no conflict of interest

Data availability

No data was used for the research described in the article.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author(s) abstained from employing any AI-assisted software.

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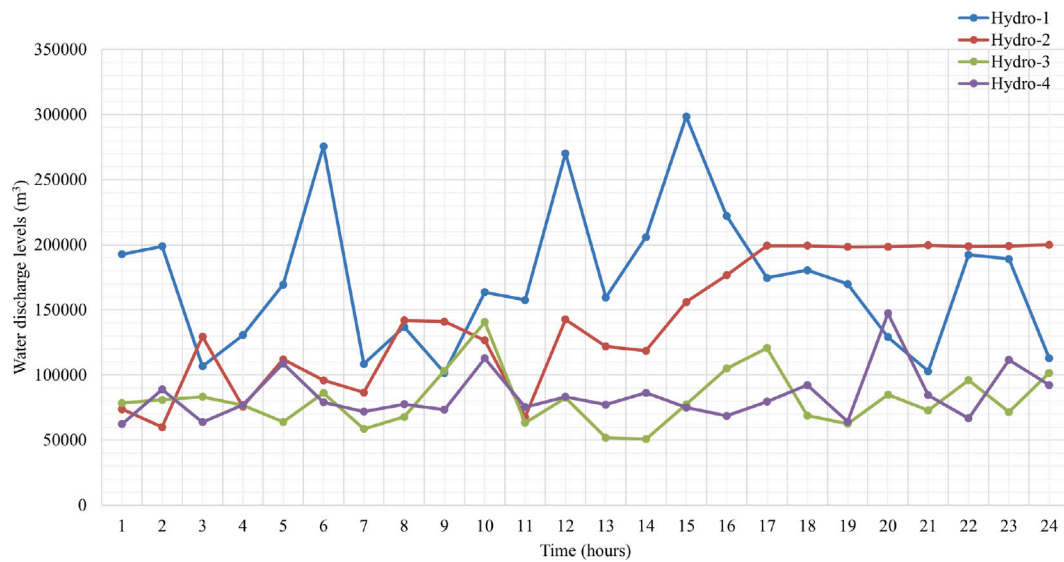


Fig. 23. Water discharge levels of hydro hubs (Case Study-III including CESs).

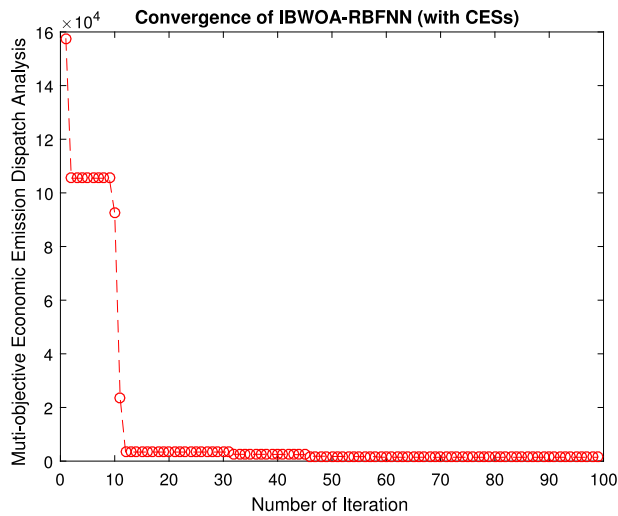


Fig. 24. IBWOA-RBFNN convergence for Case Study-III (including CESs).

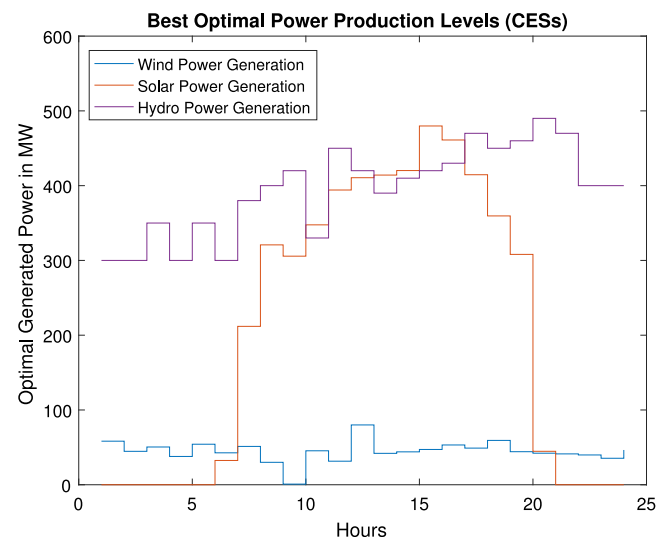


Fig. 26. Solar-wind and hydro powers levels computed by IBWOA-RBFNN (Case Study-III).

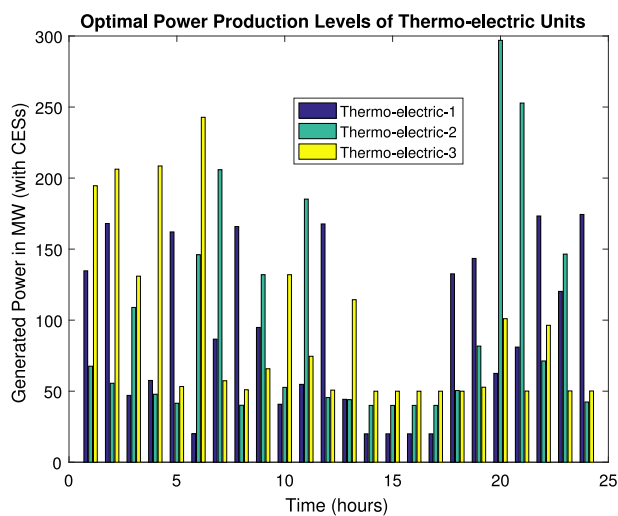


Fig. 25. Power levels of thermal hubs (Case Study-III with CESs).

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